Details and Full Results for SOLID: Spatial Optimization with Likelihood-Informed Distance Geometry for Range-Based Positioning

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This document holds the details and the full results of the paper. Numbers of sections, equations, figures, and tables that carry the prefix S belong to this document, and all others to the paper. The document keeps the notation and the names of the paper, among them the anchor-free (AF), anchor-aided cooperative (AAC), and anchor-only non-cooperative (ANC) modes. Section S-I gives the model in full and Section S-II the preparation of every dataset. Section S-III gives the full results, with the distributions on HYMN, AAU, TIERS, Bregar, and IDLab, the speed of every method, and the ablation on the training parts. Section S-IV gives the block likelihood of one link, and Section S-V the details of the comparison.

S-I. The Model in Full

This section completes the model of Sections III to VI, in their order. Section S-I-A completes the notation and the link model, and Section S-I-B explains how the block likelihood (6) is evaluated and how accurate that evaluation is. Section S-I-C completes the presence law and the heights, Section S-I-D the cooperative modes, Section S-I-E the anchor-only mode, and Section S-I-F the cost. Section S-I-G completes the fits and the fixed rules, with the settings that they give every site in Table S-I and every fixed numerical setting in Table S-III, and Section S-I-H names everything that another ranging technology would set again. The released implementation follows every rule stated here.

A. Notation, Readings, and the Block Likelihood

A hat marks an estimate, a superscript circle a true value, and a tilde a sample median, the mean of the two middle values for an even count. The reported scale and any calibration of the readings come from training or calibration readings of the site itself, so SOLID needs no error report from the radio. On TWR-CLOUD the reported scale follows the clipped linear scale law $s(y) = \min\{\max\{0.026356\,y + 4.650\,\mathrm{cm},\, 8\,\mathrm{cm}\},\, 55\,\mathrm{cm}\}$ of the reading y, whose clips lie at 127.1 and 1910.4 cm and whose fit Section S-II-A describes, and elsewhere it is one value per site or radio channel (Section S-II). The scale law follows the readings, so on TWR-CLOUD L_e is not normalized over them (assumption 5 of Appendix A). At the fitted values of TWR-CLOUD the likelihood of one reading has a mass over that reading between 0.993 and 1.026, which changes with d by at most about 0.03 nats, most of all near the clips. A reading is admissible within the bounds of Table S-III.

The residuals from which an adapter derives its reported scale carry the offsets and excesses of their links. Outside TWR-CLOUD the reported scale is their root mean square, which holds every part of their error. On TWR-CLOUD the scale law follows a robust spread of the residuals in bins of the true distance, which holds the offsets but discounts the long tail of the excesses (Section S-II-A). Across the sites the reported scale is 2.7 to 34 times the core scale. Since the offset and the excess are proportional to the median core scale of their link (Section IV-A), a constant factor common to every reported scale of a site changes only the fitted κ and leaves the block likelihood unchanged. A constant factor in the reported scales still changes the grid step, the area padding, and the score of the lateration of Section S-I-D, which read the reported scale itself (Table S-III). In the median stage the multipliers $\alpha_{\pm}$ of its scale (Section S-I-D) absorb it except in AAC on TIERS and IDLab (Section S-I-G). The fitted κ does not absorb a change in how the reported scale varies, with the reading on TWR-CLOUD or between the radio channels of Bregar.

B. Numerical Evaluation of the Block Likelihood

The block likelihood (6) has no closed form, and the estimators read it at every candidate distance, so we tabulate ℓ_e once per link over the distances

$$y_{\min} - 40\mu_e - 12\tau_e - 12\sigma_{\max} \le d \le y_{\max} + 12\tau_e + 12\sigma_{\max}, \quad \text{(S1)}$$

with $y_{\min}$ and $y_{\max}$ the smallest and largest readings and $\sigma_{\max}$ the largest core scale. The range reaches far below the readings, since an excess can place the distance there. Let

$$G_e(d+w) = \prod_{k=1}^{n_e} t_\nu\big(y_{e,k} - d - w;\, 0, \sigma_{e,k}\big) \quad \text{(S2)}$$

be the product of the densities of the individual reading errors in (2) at the shifted distance $d + w$. Each density is $t_3(\varepsilon; 0, \sigma) = 2(\pi\sqrt{3}\,\sigma)^{-1}\big(1+\varepsilon^2/(3\sigma^2)\big)^{-2}$, with the standard deviation $\sqrt{3}\,\sigma$. The curvature width $\big(\sum_k 4/(3\sigma_{e,k}^2)\big)^{-1/2}$ of G_e is a lower bound on its width. The step is the largest power of two between 2^{-11} and 2 cm that is at most 0.283 times the width of the likelihood peak. That width is τ_e when τ_e exceeds

TABLE S-I
SETTINGS AND MULTIPLIERS OF EACH SITE

Site	Prior heights ζ_q (cm)	h (cm)	Training blocks	(α_+, α_-) in AF	(α_+, α_-) in AAC
TWR-CLOUD	five, -92.0 to -32.5	4.24	2400	1.359, 1.526 (one reading)	as in AF
				0.949, 0.477 (ten readings)	as in AF
TIERS	five, 6.6 to 18.8	3.54	3864	0.446, 0.436[a,c]	0.214, 0.214[b,c]
IDLab	257.1	14.21	14 532	0.282, 0.283[a]	0.811, 0.738[b]
Bregar 0	120.0	9.12	4000		
Bregar 1	95.0	4.95	4000		
Bregar 2	132.0	17.32	4000		
Bregar 3	120.0	11.03	4000		
HYMN	five, 210.5 to 334.1	28.28	4000		
AAU	five, 46.2 to 325.0	3.54	4000		

[a]A constant of the start (Section S-I-G) with ten readings. [b]Constants of the start. [c]From the calibration recording (Section S-II-B).

a quarter of the curvature width, as at every fitted site, and the root sum of squares of the two widths otherwise. Linear interpolation of a Gaussian peak of that width then errs by at most $0.283^2/8 \approx 0.01$ nats. Where the second differences of the table predict a larger error at a point, the two steps on each side of it are tabulated again at half the step. The table becomes a sequence of runs, each with a step of its own, and the new runs are checked in the same way. The halving repeats until no point predicts more than 0.01 nats or the next step would fall below the larger of 2^{-11} cm and a sixty-fourth of the curvature width. The tolerance of 0.01 nats is therefore a target, which this floor can leave unmet. A table value is a trapezoid sum of $G_e(d+w)\, f_e(w)$ on a finer lattice of w, with f_e the shift density (5). The offset enters as a Gaussian kernel whose central weights match its mass and second moment. The excess enters as a backward recursion of the exponential convolution on long runs and as a sampled exponentially modified Gaussian on short ones.

Beyond the table, ℓ_e continues with an asymptotic form. Its core term replaces G_e by a Gaussian of equal mass, mean, and variance and convolves it with the shift in closed form. Its tail term expands $\ln G_e$ to second order at d, averages this expansion over the offset in closed form, and takes the excess through its moment generating function at the slope of $\ln G_e$. The offset average grows without bound as τ_e^2 times the curvature of $\ln G_e$ approaches one, and the moment generating function does so as μ_e times the slope approaches one. We therefore continue the offset average linearly in the product of τ_e^2 and the curvature of $\ln G_e$ once that product exceeds one half. We also replace the moment generating function by its second-order Taylor polynomial wherever the slope is positive, as below the readings. Both replacements keep the slope continuous. Where G_e stays log-concave, each term passes smoothly to a Laplace approximation. Each end of the table first grows until the table and the asymptotic form agree there within 0.05 nats, each time by the larger of half its distance from the mean of G_e and 16 table steps. The upper end also grows until the tail term exceeds the core term by 10 nats, past the distances where the tails of the reading errors take over from the offset. An end stops growing after twelve extensions, once it lies farther from the mean of G_e than eight times the width of the initial range, or once the table there has fallen 700 nats below its peak. The 0.05 nats are thus a target as well. Each extension is refined on its own, so the second differences are never taken at the point that it shares with the table. A correction that decays away from each end then gives the continuation the value and slope of the table at that end.

The modes read the table in different ways. ANC and the AAC peaks of Section S-I-D read it by linear interpolation. Beyond it they read a tabulated continuation, made of cubic Hermite pieces through the values and slopes of the continuation at points at most 4 cm apart. Every posterior and every peak search computes this tabulation anew over the slant distances that its grid can take. Most of the operating area lies beyond the table of a link, where each evaluation of the asymptotic form would cost Newton steps toward the peaks of its Laplace approximations, and the linear interpolant and the cubic pieces both admit exact bounds over an interval (Section S-I-E). The cooperative objectives read the table by cubic Hermite interpolation, which is continuously differentiable, as the limited-memory Broyden–Fletcher–Goldfarb–Shanno descent (L-BFGS) of Section V-A needs, and read the continuation itself beyond the table. The fits evaluate (6) by Gauss–Legendre quadrature on subintervals in a hyperbolic sine variable. The subintervals surround the readings, the peak of G_e, the distance d, and two points between that peak and d that carry the integrand when d lies far from the readings.

Every table value is accurate to about 10^{-3} nats. The trapezoid sum of an integrand that is analytic in a strip around the real line and decays along it converges exponentially as the lattice step shrinks, at a rate set by how far the strip reaches from the real line. The Gaussian and the exponentially modified Gaussian of the shift density (5) are analytic in the whole complex plane, so for heavy-tailed reading errors the strip reaches up to the poles of the t densities in G_e, at the points $y_{e,k} \pm \mathrm{i}\sqrt{3}\,\sigma_{e,k}$, and these poles give the leading error. For one reading and an offset much wider than its core scale σ, the relative error is at most about $2(1+\chi)\mathrm{e}^{-\chi}$, where the ratio χ of $2\pi\sqrt{3}\,\sigma$ to the lattice step is at least 10.47. A narrower offset raises the error above this value at a given step, several-fold when τ_e is as small as σ. It also refines the lattice, whose step never exceeds the table step or, at every fitted site, $1.2\,(4/(3\sigma^2) + \tau_e^{-2})^{-1/2}$. With one reading the relative error thus stays below 7×10^{-4}, which only the widest offsets approach. With several readings that cap of the lattice step becomes $1.2\,(\sum_k 4/(3\sigma_{e,k}^2) + \tau_e^{-2})^{-1/2}$, which decreases as

readings are added, while the table step depends on τ_e alone. The lattice step is the largest power-of-two fraction of the table step not above this cap, so added readings leave it unchanged until the cap falls below it. At a fixed step the error grows with the number of readings when their values coincide, since their poles then merge into poles of higher order. As the readings move apart it first falls, and once they lie many core scales apart each reading forms a peak of its own, whose numerical error returns toward that for one reading at the same step. Even at the cap the numerical error for coinciding readings stays below the bound for one reading.

Between the table values, the cubic Hermite interpolant is exact for a Gaussian peak. Its slopes are differences of the table values, so otherwise it adds an error of third order in the step. Between interior points of a run this error is at most about $0.016\,|\ell_e'''|$ times the cube of the step, and in the steps at the ends of the table and beside a change of step it is up to four times that. On 74 synthetic blocks of one, four, or ten readings, on 27 parameter sets, none of them a fitted set of Table III, both interpolants agreed with an independent adaptive quadrature and with the quadrature of the fits within 0.02 nats inside the initial range of the table, before its ends grow, at 25 to 38 sampled distances per block. Beyond that range the interpolants, and the continuation beyond the table, agreed with the two quadratures within 0.1 nats at 17 distances on each side, fourteen of them from 2 cm to 10 m out and three at a hundredth, a tenth, and the whole of the length of the initial range, which is 4 to 114 m. Each bound allowed a further $2 \times 10^{-6}|\ell_e|$, below 6×10^{-4} nats in these blocks. Between and beyond these distances, out to 100 km, the continuation decayed away from the table without rising above its peak. Far beyond the table the ANC refinement uses the continuation to bound whole quadtree squares and drops the squares far from the agent (Fig. 2).

C. The Presence Law and the Heights

The frame graph $\mathcal{F}$ has a vertex for each agent, one frame vertex $\star$ that stands for all anchors, and an edge for every success, and we write $i \leftrightarrow j$ when a path of $\mathcal{F}$ joins i and j. Every agent attempts every radio that its mode ranges (assumption 6), so the opportunities of Section IV-B are

$$\mathcal{O} = \mathcal{O}_{\mathcal{V}} \cup \big\{\{i, a\} :\ i \leftrightarrow \star,\ a \in \mathcal{A}_{\mathrm{act}}\big\}, \tag{S3}$$

with $\mathcal{O}_{\mathcal{V}} = \{\{i, j\} \subseteq \mathcal{V} : i \leftrightarrow j\}$ in the cooperative modes and empty in ANC, $\mathcal{A}_{\mathrm{act}}$ the anchors that some agent heard, and $\mathcal{O}_i$ the anchors a with $\{i, a\} \in \mathcal{O}$. In the cooperative modes a failure between two components of $\mathcal{F}$, groups of agents that no chain of successes joins, would cost only 10^{-12} nats once the components move apart to the bound of (7). In ANC such a failure belongs to an agent that heard no anchor, which cannot move apart from the anchors within the operating area (assumption 8), so assumption 6 alone leaves it out.

In the coordinate frame of the anchors of TWR-CLOUD, whose origin lies above every radio, the agent heights of the training part run from -93.7 to -10.6 cm, the span of 83.1 cm of Section IV-C. On TIERS and IDLab, the two other sites where agents range each other, the heights of the radios span 13.6 and 9.5 cm, so the gap between the slant and the horizontal distance of an agent pair reaches at most 0.92 and 0.45 cm at 1 m and less beyond.

The ten assumptions of Appendix A have consequences that the list leaves implicit. The error terms $\varepsilon_{e,k}$ in assumption 1 are independent across readings and directions. Assumption 3 takes the admissible readings of a measured link to follow L_e alone, so their density given a success, their reported scales, and the distance d is $L_e(d)$. Assumption 5 concerns the number and the values of the reported scales of a success.

D. The Cooperative Modes

The median stage of Section V-A replaces the block term of link e by a term in its median reading $\tilde{y}_e$,

$$\rho_e^{\mathrm{med}}(d) = \rho_\delta\!\left(\frac{\tilde{y}_e - d}{\alpha_\pm\, s_e^{\mathrm{med}}(d)}\right) + \ln s_e^{\mathrm{med}}(d), \tag{S4}$$

with α_+ where $\tilde{y}_e \geq d$ and α_- otherwise, set per site in Section S-I-G. The scale $s_e^{\mathrm{med}}(d)$ is the scale law $s(d)$ on TWR-CLOUD. Elsewhere it is the reported scale of the middle reading in sorted order in AF, or the mean of the two middle ones, and in AAC the median of the reported scales. On TIERS and IDLab both equal the one scale that every reading of a link carries. The pseudo-Huber loss $\rho_\delta(\eta) = \delta^2(\sqrt{1 + \eta^2/\delta^2} - 1)$ is quadratic for small residuals and linear beyond about $\delta = 0.7$. The pull $|\rho_\delta'|$ of one standardized median residual rises toward δ and stays below it. The standardized residual falls as d grows, and on TWR-CLOUD $\ln s(d)$ rises only slowly, so on every site a term of (S4) has no minimum farther than 2.7 cm from its median. A stray reading can move that minimum but never splits it into minima more than 2.7 cm apart, whereas it sets the peak of the block likelihood of a link with one reading and splits that of a link with two, such as a TWR-CLOUD agent pair with one reading in each direction. Where the scale of (S4) is constant, as on TIERS and IDLab, a term has its only minimum at the median, and its slope approaches $\delta/(\alpha_\pm s_e^{\mathrm{med}})$ far from the median. A link then keeps a pull of nearly fixed size where the heavy tails of the reading errors let the slope of the block term fade, and a layout far from the truth still feels every link. By the same bound, one median far from its distance moves the start layout little there.

On TWR-CLOUD the scale $s_e^{\mathrm{med}}(d) = s(d)$ of (S4) follows the candidate distance between the clips of the scale law (Section S-I-A), and the two sides of the median differ. Between the clips $\ln s(d)$ rises at the rate $0.026356/s(d)$, at most 0.0033 nats/cm. This rate moves the minimum of a term below a median between the clips by about $\alpha_+^2(0.026356)\, s(\tilde{y}_e)$, which reaches the bound of 2.7 cm at $\alpha_+ = 1.359$ and a scale of 55 cm. No term has a minimum above its median. A median less than 2.7 cm above the upper clip can have a second minimum within that distance, behind a ridge of less than 10^{-3} nats.

Below the median reading the slope of a term grows as d falls between the clips. The part of the slope that ρ_δ contributes there is $(0.026356\,\tilde{y}_e + 4.650\,\mathrm{cm})/s(d)$ times its value under a scale held at $s(d)$. A median $\tilde{y}_e$ far longer than a candidate

distance d, both between the clips, thus gives the term a slope whose size is about $\delta\, s(\tilde{y}_e)/(\alpha_+\, s(d)^2)$. This is $s(\tilde{y}_e)/s(d)$ times the slope $\delta/(\alpha_+\, s(d))$ under a scale held at $s(d)$, and $(s(\tilde{y}_e)/s(d))^2$ times the slope $\delta/(\alpha_+\, s(\tilde{y}_e))$ under a scale held at $s(\tilde{y}_e)$, the scale of the median reading itself. A link that a layout makes too short thus pulls ever harder between the clips, but so does a median far longer than its distance. Below the lower clip the scale law stays at 8 cm, so there the slope is $\delta/(8\,\mathrm{cm}\,\alpha_+)$, as under a constant scale of 8 cm, and $s(\tilde{y}_e)/(8\,\mathrm{cm})$ times the slope under $s(\tilde{y}_e)$. Where d far exceeds the median reading, the standardized residual levels off toward $-1/(0.026356\,\alpha_-)$, so $|\rho'_\delta|$ stays near δ. The slope of the term itself falls roughly as $\delta\,(0.026356\,\tilde{y}_e + 4.650\,\mathrm{cm})/(\alpha_-\, s(d)^2) + 0.026356/s(d)$, a gradual fade. Past its largest value the slope of the block term instead falls steeply toward $4n_e/(d-\tilde{y}_e)$, the slope of the tails of the reading errors. Beyond the upper clip the scale stays at 55 cm, and the slope of the term settles near $\delta/(55\,\mathrm{cm}\,\alpha_-)$. A link that a layout makes far too long is still felt, and a median far shorter than its distance moves the start layout little.

With $C_\delta = 2\delta \mathrm{e}^{\delta^2} K_1(\delta^2) = 3.884$, K_1 the modified Bessel function of the second kind,

$$p^{\mathrm{med}}(\tilde{y}_e \mid d) = \frac{2\exp\bigl(-\rho_e^{\mathrm{med}}(d)\bigr)}{C_\delta\,(\alpha_+ + \alpha_-)} \tag{S5}$$

is a normalized density of $\tilde{y}_e$ given d. The median stage is thus a maximum likelihood problem for the median readings and the outcomes,

$$J_{\mathrm{med}}(\mathbf{X}) = \sum_{e\in\mathcal{E}} \rho_e^{\mathrm{med}}\bigl(d_e^{\mathrm{hor}}\bigr) - \sum_{e\in\mathcal{O}} \ln P_{\hat{r}}\bigl(o_e \mid d_e^{\mathrm{hor}}\bigr), \tag{S6}$$

with every pair at its horizontal distance d_e^{hor}, which is $d_{ij}(\mathbf{X})$ or $\|\mathbf{x}_i - \mathbf{p}_a\|$. The median-stage objective J_{med} has kinks at the clips of the scale law of TWR-CLOUD. The tables of ℓ_e fix the scale of each reading at its observed value, so J has no kink at the clips. Both objectives have kinks at the bounds of (7) and where two positions coincide.

The range-only start layout $\mathbf{X}^{\mathrm{ro}}$ of Section V-A follows from the readings of the scenario by deterministic rules. They read the direction median $\tilde{y}_{\vec{e}}$ of one direction $\vec{e}$ of link e, the median of the readings of that direction as the adapter delivers them. The pair median of an agent pair is the mean of its two direction medians when both directions hold admissible readings, and the one direction median otherwise. TWR-CLOUD delivers the two directions of an agent pair apart, so with ten readings per direction the pair median differs from the median reading $\tilde{y}_e$ of the pooled readings. With one reading per direction, and on TIERS and IDLab, which do not deliver the two directions of a pair apart, the two agree. In AF, $\mathbf{X}^{\mathrm{ro}}$ is the classical multidimensional scaling (Section V-A) of the shortest-path distances between the agents along the agent-pair links, each link entering with its pair median as its length, so a linked pair enters at the shorter of its pair median and the shortest path through other agents. Each component of the links is embedded on its own, and the centroids of successive components lie apart along the first axis by the spacing of Table S-III.

In AAC, $\mathbf{X}^{\mathrm{ro}}$ aligns each component of that layout by one rigid motion, reflection allowed, of least squared error to the targets of its agents. The target of an agent is its lateration, defined below, or, where its anchor links fix its position, the peak of $\sum_q \exp\Lambda_{i,q}(\mathbf{x};\varnothing)$, at which that agent then sits. An agent without anchor links has no target and moves with its component. With laterations as the only targets, so that no agent sits at a peak, the same alignment gives the initializer layout of SOLID (Section S-V-B), which equals $\mathbf{X}^{\mathrm{ro}}$ in AF. In ANC that layout places every agent with an anchor link at its lateration and every other agent at its starting coordinates (Section S-II). The anchor links of an agent fix its position when they reach at least three distinct horizontal anchor positions that do not all lie near one line (Table S-III). Such an agent sits at a maximizer of the sum on the grid of Table S-III over the bounding box of the squares around its linked anchors, each with a half side equal to the upper end of the table of its link. A branch and bound with the bounds of Section S-I-E finds it level by level. It bounds every quadtree square of one level, drops each square whose upper bound falls below the largest value certified so far, and splits the others until single cells remain. The peak is exact unless a level exceeds the memory guard of Table S-III.

The lateration of an agent places it from the direction medians of its anchor links alone. It takes the height of the agent to be the median height $\tilde{z}$ of the anchors of the scenario. It turns the direction median $\tilde{y}_{\vec{e}}$ of each direction of its link to anchor a into the horizontal radius $(\max\{\tilde{y}_{\vec{e}}^{\,2} - (z_a - \tilde{z})^2, 0\})^{1/2}$ around $\mathbf{p}_a$ and takes the median radius per anchor. With one anchor the lateration is the point at that radius from $\mathbf{p}_a$ along the first coordinate axis. With two anchors, taken in a fixed order, it is the intersection of the two circles to the left of the line from the first anchor to the second. Where the circles do not meet, it is the point of that line at which the squared distances to the two anchors differ as the squared radii do. With three or more it is the candidate of least score among the linear least-squares laterations from all anchors and from every three anchors, each of which subtracts the circle equation of one anchor from those of the others. The score sums $\rho_\delta((\tilde{y}_{\vec{e}} - d_{ia}(\mathbf{x},\tilde{z}))/s)$ over the direction medians of the agent, with $\delta = 0.7$ as in (S4) but without the multipliers $\alpha_\pm$, and with s the reported scale of the direction median, $s(\tilde{y}_{\vec{e}})$ on TWR-CLOUD.

Of the 64 starts of a stage (Section V-A), small displacements of the start layout probe its basin, and large ones can reach another basin. Start $g = 1, \ldots, 63$ adds $d_{\mathrm{typ}}\gamma_g \boldsymbol{\xi}_g$ to the coordinates of the start layout. The amplitude γ_g cycles through the six values of Table S-III, and each coordinate of $\boldsymbol{\xi}_g \in [-1,1)^{2N}$ is a fixed pseudo-random number of the start and the coordinate. The typical distance d_{typ} is the median of the median readings $\tilde{y}_e$ over the links in AF and their upper middle value in AAC, and at least 1 cm, and in a scenario without links it is 100 cm. The rigid alignment that builds $\mathbf{X}^{\mathrm{ro}}$ in AAC can move an agent away from a good lateration of its own, so the added start of AAC moves every agent linked to three or more anchors to its lateration and keeps the other agents where the stage starts. A stage leaves this start out when it would repeat the start layout, that is, when no

agent is linked to three or more anchors or every such agent already sits at its lateration. In AF the start layout $\mathbf{X}^{\mathrm{start}}$ of the median stage is $\mathbf{X}^{\mathrm{ro}}$. In AAC it is built as $\mathbf{X}^{\mathrm{ro}}$ is, with the peaks taken under $\hat{r}$, so a change of a peak can move every agent of its component. With $\mathrm{MS}(J, \mathbf{X})$ the multistart of J from $\mathbf{X}$, the estimate is

$$\hat{\mathbf{X}} = \mathrm{MS}\big(J,\, \mathrm{MS}(J_{\mathrm{med}}, \mathbf{X}^{\mathrm{start}})\big), \tag{S7}$$

so the block stage starts from the best layout of the median stage, and its best run is the estimate.

Each iteration of L-BFGS takes the standard two-loop direction from the stored steps and their gradient changes, with its initial matrix scaled by the newest of them. Its line search accepts the first of the trial steps of Table S-III that gives the sufficient decrease named there, a share of the decrease that the slope at the start predicts, without the curvature condition of the Wolfe line search. An accepted step can therefore have nonpositive curvature, and a step enters the memory only when its inner product with its gradient change exceeds 10^{-10} times the product of their norms. A direction whose slope is not both finite and negative is replaced by the steepest descent direction with an emptied memory. No run increases its objective. The objectives are only piecewise smooth and the line search asks only for a sufficient decrease, so a run need not end at a stationary point.

The components of a cooperative objective are those of the frame graph $\mathcal{F}$ when $\hat{r} \neq \varnothing$, since any two agents that $\mathcal{F}$ joins form an opportunity, and those of the links $\mathcal{E}_{\mathcal{V}}$ when $\hat{r} = \varnothing$. In AF each component keeps its own rotation, reflection, and translation. In AAC a rigid motion of a component leaves J unchanged exactly when it keeps the horizontal positions of the anchors in its terms, and anchors stacked at one horizontal position count once. These anchors are all of $\mathcal{A}_{\mathrm{act}}$ for the component of $\mathcal{F}$ that holds $\star$ when $\hat{r} \neq \varnothing$, and otherwise the anchors linked to its agents. Three or more such positions that do not lie on one line fix the frame of the component. Two or more positions on one line leave the reflection across that line, and a single position leaves every rotation about it and every reflection across a line through it. A component with no anchor in its terms keeps the freedom of AF. With $\hat{r} = \varnothing$ the links can also leave a component free to bend, as when an agent whose only link goes to another agent turns about that agent.

E. The Anchor-Only Mode

Through $\mathcal{A}_{\mathrm{act}}$, which the agents of ANC share, the failures of an agent depend on what the others heard. Outside TWR-CLOUD the anchor readings of all agents and their reported scales set the operating area $\mathcal{X}$ (Section S-I-G). Through $\mathcal{X}$ the grid, the reach integrals, and the draws of an agent then depend on the readings of the others. An agent with neither a link nor a presence term keeps the uniform prior over $\mathcal{X}$.

The cells $c \in \mathcal{G}$, at whose centers $\mathbf{x}_c$ Appendix B evaluates $\psi_i(c)$, are squares of side h laid from the lower corner of $\mathcal{X}$. Along each axis the first center lies half a cell inside the lower bound of $\mathcal{X}$, and the next ones follow at the step h for as long as they stay below its upper bound. The cells therefore end within half a cell of each upper edge of $\mathcal{X}$, short of it or beyond it, and the sums over the cells and the draws of this subsection cover their union in place of $\mathcal{X}$. The bounds of the base grid step (S10) bind on TIERS and AAU at 5 cm and on HYMN at 40 cm. At $\tilde{s}$ a cell spans 0.24 offset standard deviations τ_e on TWR-CLOUD, three on AAU, and 0.37 to 1.15 elsewhere. On AAU the grid posterior is thus a coarse midpoint approximation.

The bounds $\psi^{\mathrm{lo}} \leq \psi_i(c) \leq \psi^{\mathrm{up}}$ of Appendix B carry fixed allowances for rounding, whose sizes Table S-III gives. Every bound of a term and each bound of the log-sum over the prior heights is widened for the rounding of the few operations behind it and of the last bits of the exponential and the logarithm, which need not be monotone. A failure near the bound $1 - 10^{-12}$ of (7) needs more, since the last bit of the exponential lets a farther cell receive a success probability up to 2^{-51} of its value above that of the near end of the interval, and the presence term $\ln P_r(0 \mid d)$ of a failure magnifies this error by $P_r(1 \mid d)/P_r(0 \mid d)$, up to 10^{12}. Both bounds of the presence term of a failure are therefore widened by a further allowance of twice this magnified error at the near end d. The bounds thus hold for the cell values that the grid computes with the interpolated table and its tabulated continuation. The running sums of the bounds on the kept and the dropped mass are not widened, which is why Appendix B states its certificate in exact arithmetic.

A square that the refinement keeps whole (Section V-B) is a leaf, and a leaf of several cells is a coarse leaf. A draw first picks a leaf by its weight, then a cell of the leaf by rejection, accepted with probability at least e^{-1} in a coarse leaf, and then a uniform point in that cell.

ANC judges the reach by (21) of Appendix C, a log marginal likelihood that on TWR-CLOUD is that of the working likelihood. The values $\lambda(r)$ come from the posteriors without presence terms, in one pass over their kept cells for all nine candidates. The running sums over the cells and the agents are not widened either, and the relative allowance with which Appendix C excludes a candidate is assumed to cover their rounding. Each candidate that the test of Appendix C leaves open is then refined under its own law, by the posterior of every agent under that candidate, whose kept mass and bound on the dropped mass narrow the bounds of λ there before the test runs again. Candidates that still stay open go to an uncertified fallback, which takes the one of largest lower bound when that bound, without the margin of what the denominators may miss, exceeds λ_0 and otherwise sets $\hat{r} = \varnothing$.

The risks (13) and (14) depend on $\mathbf{X}$ only through pair distances, so their minimizers form a union of orbits under rotations, reflections, and translations, and they fix only the pair distances of a layout. In (14) every sample caps the pair errors at its own diameter $D(\mathbf{X}^{(l)})$. Each of the two candidates of $\hat{\mathbf{X}}$ descends along the normalized gradient of $\hat{R}$, which exists almost everywhere, and keeps a step only when $\hat{R}$ falls (Table S-III). In the gradient every pair pushes its two agents by equal and opposite amounts along the line between them, so each step keeps the centroid of the layout and does not turn it as a whole. The reported layout thus keeps the centroid of its candidate exactly and its orientation up to a small drift

of second order in the step lengths. The two candidates need not share a centroid, and which one is kept can turn on the Monte Carlo error of $\hat{R}$, so $\hat{\mathbf{X}}$ is a relative layout whose pair distances, not its placement, form the estimate.

With two agents the cap of (1) is the pair distance of the second layout itself, the sampled pair distance in (14). Against a sampled pair distance d', a layout with pair distance d thus loses $\min\{|d - d'|, d'\}$, and the posterior risk of (13) has the derivative $2H(d) - H(d/2) - 1$ in d, with H the posterior distribution function of the pair distance. The pair distance d^{B} of the Bayes decision $\mathbf{X}^{\mathrm{B}}$ therefore solves $H(d^{\mathrm{B}}) = (1 + H(d^{\mathrm{B}}/2))/2$, and it lies above the posterior median whenever the posterior puts mass below $d^{\mathrm{B}}/2$. The descent follows the same derivative with the distribution function of the M sampled pair distances in place of H, but it need not reach a minimizer of (14), so the pair distance of the reported layout $\hat{\mathbf{X}}$ approximates d^{B} without a certificate.

The edge cases of ANC are these. A single agent, for which (1) is not defined, reports its posterior mean. When a posterior that the memory guard cut keeps no cell, its agent draws uniformly over every cell of $\mathcal{G}$, as an agent with neither a link nor a presence term does. Without a cut, a posterior keeps no cell only when it is finite on no cell, and ANC then returns no layout for the scenario. A Bregar scenario without any anchor reading has no operating area and gets no estimate.

Table S-II follows the refinement through the scenario of Fig. 2, training scenario 10024 of the first scan of TWR-CLOUD, solved with the fitted values and settings of TWR-CLOUD (Tables III and S-I) on readings that the range line of the other ten training scans calibrated (Section S-II-A). Its four agents heard seven of the eight anchors between them, so each agent has seven opportunities, its links and its failed opportunities, whose outcomes enter at the reach $\hat{r} = 1098.9$ cm that the scenario chose. In Table S-II the kept cells of an agent include the cells of its coarse leaves, and its bounded squares are the quadtree squares whose bounds the refinement computed, both for its posterior under $\hat{r}$. Agents 19 and 38, linked to four and five anchors, kept almost only single cells near one compact peak. Agent 122, linked to three anchors, kept its cells along the elongated peak of Fig. 2, while agent 101, linked to two anchors, kept the most, three quarters of them in coarse leaves that the refinement never split. Over every posterior of the scenario, those without presence terms that judge the reach included, the refinement bounded 12 888 quadtree squares and evaluated 38 849 cells, less than 15 percent of the cells of the grid.

F. Cost

Every cost below is analytical. One evaluation of a cooperative objective visits every link and every opportunity, at most $N(N-1)/2$ agent pairs and $N|\mathcal{A}_{\mathrm{act}}|$ pairs of an agent and an anchor, and reads Q table values for each anchor link of each agent, so its cost grows quadratically with the number of agents. Classical multidimensional scaling with its shortest paths costs $O(N^3)$. For the AAC starts, each agent linked to three or more anchors needs the laterations of all triples of its anchors, each scored over its direction medians, $O(|\mathcal{A}_i|^4)$ operations in all. Each agent whose anchor links fix its position also needs one peak search for $\mathbf{X}^{\mathrm{ro}}$ and, when $\hat{r} \neq \varnothing$, a second one under $\hat{r}$ for $\mathbf{X}^{\mathrm{start}}$. Each stage runs 64 starts in AF and up to 65 in AAC, each of at most 10^4 iterations with up to 30 trial steps. An ANC posterior costs $O(Q(|\mathcal{A}_i| + |\mathcal{A}_{\mathrm{act}}|))$ per evaluated cell, since $\mathcal{O}_i = \mathcal{A}_{\mathrm{act}}$ for every agent that heard an anchor. When some opportunity fails, each such agent also gets a posterior without presence terms for the reach and at most nine posteriors under the open candidates, so ANC computes at most eleven posteriors per agent. The descent toward $\mathbf{X}^{\mathrm{B}}$ stores $MN(N-1)/2$ sample pair distances of eight bytes, 1.4 MB at 19 agents and 4.1 GB at 1000, and passes over them at most 532 times.

G. Fitting the Model to a Site

The training blocks of (15) take the slant distance as their true distance d°_{β}, except the agent pairs of TWR-CLOUD, which enter at the horizontal distance of the cooperative modes (Section IV-C). On every site but TWR-CLOUD the one-reading block of a link repeats the first reading of its ten-reading block, and on TWR-CLOUD the two are separate draws from the same recorded readings of the link. On TWR-CLOUD the mass of L_e over the readings varies with $\boldsymbol{\theta}$ as well as with d, and Section S-I-A gives its size at the fitted values. Each fit of (15) keeps the best of 15 Nelder–Mead searches within the box Θ, each polished by the bound-constrained L-BFGS (L-BFGS-B), with the settings of Table S-III, and all nine fits gained less than 0.01 nats at their first restart. On IDLab every link setting, a pair of radios under one radio configuration, carries equal weight within a reading count. The fits keep only admissible readings and do not renormalize L_e to the admissible interval, which matters only for a link within a few offset standard deviations of an end of that interval. In Table S-I, rounded for print, the rows Bregar 0 to 3 are the four Bregar environments, the prior heights are absolute heights in the coordinate frame of the anchors of each site, and only the datasets of the cooperative modes carry multipliers.

We fit the multipliers of (S4) by weighted maximum likelihood of the normalized density (S5) to standardized residuals $\eta \in \mathcal{Z}$, each a reading, or a median of readings, less its true distance and divided by a scale,

$$(\hat{\alpha}_+, \hat{\alpha}_-) = \operatorname*{arg\,min}_{\alpha_+, \alpha_-} \sum_{\eta \in \mathcal{Z}} \omega_\eta^{\mathrm{med}} \rho_\delta\big(\eta/\alpha_{\pm}\big) + \ln(\alpha_+ + \alpha_-), \quad \text{(S8)}$$

with α_+ where $\eta \geq 0$ and α_- otherwise, and weights $\omega_\eta^{\mathrm{med}}$ that sum to one. The problem is convex in $(\ln \alpha_+, \ln \alpha_-)$. On TWR-CLOUD the residuals are medians of readings drawn as a scenario draws them, each less the horizontal true distance d° and divided by $s(d^{\circ})$. Every pair of radios of a training scan enters both as an agent pair with its two directions pooled and as two anchor links of one direction each, the two kinds with half of the weight each, and one pair of multipliers per reading count serves both cooperative modes. The pair of a reading count thus estimates the law that the median stage of AAC applies, with the two kinds in equal shares rather than in the shares of a scenario. The median stage of AF scores agent

TABLE S-II
THE FOUR AGENTS OF THE SCENARIO OF FIG. 2

Agent	Links	Failed opportunities	Kept cells	Coarse leaves, their cells	Bounded squares	Bound on the dropped share
19	4	3	1431	21, 84	723	2.5×10^{-9}
38	5	2	1375	13, 52	697	1.7×10^{-9}
101	2	5	9998	1472, 7544	3109	7.8×10^{-10}
122	3	4	4943	523, 2104	2079	3.6×10^{-9}

pairs only, so half of the weight behind its multipliers rests on one-direction medians that AF never scores. On TIERS and IDLab the residuals are single readings at their true distances, from the calibration recording on TIERS (Section S-II-B) and from the training part on IDLab, and one pair per mode serves both reading counts. The AF fit divides each residual by the constant reported scale, as the median stage does, so with one reading its multipliers estimate the law that the stage applies, while with ten readings the stage applies them to medians, whose law no fit estimated. The AAC fit divides the residuals by a scale that grows with the distance (Table S-III), while the stage divides by the constant reported scale. The AF multipliers of these two sites with ten readings and their AAC multipliers are therefore constants of the start, numbers that only shape where the block stage begins, rather than estimates of the law that the median stage applies. The medians of one link of TWR-CLOUD share its offset and excess, as do the readings that TIERS records at one distance and orientation and the readings of one IDLab link setting, so (S8) is a composite likelihood as well. It has a unique minimum when residuals of both signs occur.

Let $z_{0.05}$, $z_{0.5}$, and $z_{0.95}$ be the linearly interpolated percentiles of the true heights of the distinct agent positions in the training part. The prior heights of AAC and ANC are

$$\zeta_q = z_{0.05} + \tfrac{q-1}{4}\big(z_{0.95} - z_{0.05}\big), \qquad q = 1, \ldots, 5, \qquad \text{(S9)}$$

five heights evenly spaced from the 5th to the 95th percentile and rounded to 0.1 cm, or the single prior height $z_{0.5}$ where $z_{0.95} - z_{0.05}$ is below 10 cm. Let $\tilde{s}$ be the median reported scale of the anchor readings in the training scenarios of both reading counts, each reading counted in every scenario that carries it. The base grid step is a quarter of $\tilde{s}$, held between 5 and 40 cm,

$$h_0 = \min\big\{40\,\text{cm},\ \max\{5\,\text{cm},\ \tilde{s}/4\}\big\}, \qquad \text{(S10)}$$

rounded to 0.1 cm. A budget factor of two, the same for every site, sets the grid step of ANC to $h = h_0/\sqrt{2}$, which halves the area of a cell, and doubles a base budget of 512 joint samples to the $M = 1024$ of Section VI. The step is a numerical budget set from the reported anchor scale, not from the width of the block likelihood, which κs and τ_e set (Section S-I-E).

Outside TWR-CLOUD the operating area $\mathcal{X}$ (Section VI) is built from the box of the anchors padded by the largest anchor reading plus three times the largest reported anchor scale, which holds every agent whose anchor readings are not grossly short. A published extent is a width and a depth that do not place the site in the coordinate frame of the anchors, but every point of the site lies within the diagonal of the site from each anchor on it. Where a site publishes its extent (Section S-II), the padded box is therefore cut to the points whose coordinates lie within that diagonal from those of every anchor, which keeps the whole site wherever it lies. TIERS, IDLab, and AAU publish theirs, and a scenario of these sites in which some agent has no anchor reading takes the points of that cut alone, without the padded box. A HYMN scenario of that kind takes the anchor box padded by 59.96 m, the longest reading that its adapter admits. On TWR-CLOUD one area serves every scenario, the box of the radio positions of the training scans, widened on each side by 5 percent of its width and depth, 64.3 cm and 152.8 cm, which measures 14.1 m by 33.6 m. Every agent of the training parts lies in its operating area, on TWR-CLOUD by construction, since the area built in the same way from the radio positions of any ten training scans equals the full area.

H. Independence From the Ranging Technology

Section VI fits or sets by fixed rules what is specific to a site and its ranging technology. The block-stage objectives and the ANC posterior read the readings and outcomes of a scenario only through L_e and P_r. The steps that prepare them also read the measurements through these two laws, in the AAC peaks, the reach values (20) and (21), and the median stage. Beyond these laws they read the readings only as median readings and reported scales, and outside TWR-CLOUD the operating area of Section S-I-G also reads the largest anchor reading. The readings also enter the seed of the draws (Table S-III), which decides which points a posterior yields, not the posterior, so no evidence enters through it. By design, the form of the estimators therefore does not change with the ranging technology. The block likelihood of another error law would enter them in the same way once it has a table and a continuation of its own, which Section S-I-B builds for (6) only.

The joint model does depend on the technology. The form of the link model (4) follows ranging by the propagation time of the first path. A technology that also ranges by propagation time would bring its own adapter, with its calibration, its scale law, and its bounds of an admissible reading. It would rerun the fits and fixed rules of Section VI on the training part of each of its sites and set again the four settings that Section VI keeps the same for every site. The presence law has nothing else to refit, since (8) estimates the reach per scenario and its candidates (18) scale with the median readings. The fixed numerical settings that are lengths or multiples of the reported scale may need new values, and Table S-III marks them, among them the three that read the reported scale itself rather than a fitted multiple of it (Section S-I-A). Ultra-wideband two-way ranging is the evidence of the paper and the only technology that tests the form (4), on the six datasets of Section S-II.

S-II. DATASET PREPARATION

This section gives for each dataset the rule that splits it into a training part and an evaluation part, the adapter that turns its reports, the recorded exchanges of Section III, into range readings, and the scenarios that it poses, so that the results

TABLE S-III
FIXED NUMERICAL SETTINGS OF SOLID

Setting	Value	Reason
Admissible reading†	reading in $[0, 10^5]$ cm, scale in $(0, 10^5]$ cm, TWR-CLOUD time of flight in $[0, 200]$ ns, and HYMN and AAU reading in $[0, 59.96]$ m	keeps corrupt readings out of every likelihood, far beyond indoor distances
Bounds of (7)	10^{-12} for a success and for a failure, acting beyond $7.43\,r$ and below $1.4 \times 10^{-6}\,r$	act like a small chance that an outcome does not follow distance and cap the evidence of one outcome at 27.6 nats
Reach candidates (18)†	nine, from $y_{0.9}/4$ to $4y_{0.9}$ in factors of $\sqrt{2}$, at least 1 cm	span from well inside to well beyond the longest links
Reach quantile	0.9	a high rank that the longest tenth of the median readings cannot raise once there are ten or more
Pseudo-Huber loss (S4)†	$\delta = 0.7$, also in the score of the lateration	bounds the pull $\lvert\rho'_\delta\rvert$ of one standardized median below 0.7 (Section S-I-D)
Displacements of the starts†	amplitudes γ_g of 0.03, 0.07, 0.15, 0.30, 0.55, and 0.90, cycled, in units of the typical distance d_{typ} of Section S-I-D	from the basin of the start layout to the size of the layout
Peak rule of the AAC start†	at least three distinct horizontal anchor positions that no strip narrower than 200 cm contains, 1 cm grid	fewer positions, or positions near one line, leave a ring or two mirror-image peaks
Component spacing†	100 cm between the centroids of successive components along the first axis	an arbitrary spacing that gives each component, and each agent without links, a center of its own (Section S-I-D)
L-BFGS†	ten steps in memory, trial steps 1 to 2^{-29} by halving until a sufficient decrease of 10^{-4}, no curvature condition, stop at a largest gradient component of 10^{-5} nats/cm, at 10^4 iterations, or without an accepted step	every accepted step lowers the objective, and only steps of positive curvature enter the memory, so every direction descends
Table of ℓ_e†	range, step, and refinement of Section S-I-B, targets of 0.01 nats for the interpolation and 0.05 nats at the ends of the table	aim at a small interpolation error and a small mismatch at each end, within a capped table size
Dropped share	10^{-4} of the grid mass	a negligible mass, left out with a certified bound
Leaf gap	1 nat between the bounds of a quadtree square kept whole as a leaf	exact draws with acceptance at least e^{-1}
Memory guard	2^{23} quadtree squares	bounds the memory of one posterior or one peak search
Rounding allowances	every bound of a term widened by 2^{-40} times one plus its size, each bound of the log-sum over the prior heights by 2^{-40} times one plus its size per prior height, both bounds of the presence term of a failure by a further $2^{-50} P_r(1 \mid d)/P_r(0 \mid d)$ at the near end d, at most 8.9×10^{-4} nats, and the reach comparisons by a relative 2^{-30}	allow for rounding, which a failure near the bound of (7) magnifies (Section S-I-E)
Seed of the draws	a fixed seed, mixed with the agent identifiers, the positions and heights of the linked anchors, and the readings of the links	random numbers of its own for every scenario, the same draws, and so the same estimate, in every run of it
Least-risk sample	the joint sample of least $\hat{R}$ among 49, every 21st	a second candidate that is itself a plausible layout
Descent toward $\mathbf{X}^{\mathrm{B}}$ (Section S-I-E)†	first step 0.05 times the larger of 1 cm and the upper middle value of the 1024 sample diameters, factor 0.6, stop below 0.01 cm or after 120 iterations	a finite refinement that never raises the risk
Box Θ of (15)	$\kappa, \mu_{\mathrm{rel}} \in [10^{-3}, 10^3]$, $\pi_{\mathrm{b}} \in [10^{-6}, 1 - 10^{-6}]$, $\tau_{\mathrm{rel}} \in [10^{-4}, 10^3]$	keeps every link parameter finite, and is reached on Bregar 0 and AAU (Appendix D)
Initial values of (15)	15, π_{b} of 0.05, 0.25, 0.5, 0.8, and 0.95, each with $(\kappa, \mu_{\mathrm{rel}}, \tau_{\mathrm{rel}})$ at $(1, 1, 1)$, $(\frac{1}{3}, 3, 3)$, and $(3, \frac{1}{3}, \frac{1}{3})$ times moment estimates, κ from the spread of the readings in blocks of three or more readings, within $[0.01, 100]$ or 0.3 without such a block, and $\mu_{\mathrm{rel}} = \tau_{\mathrm{rel}}$ from the spread of the block medians, within $[0.3, 30]$	the fit is not concave, and the initial values span a rare to an almost certain excess and trade the core scale against the offset
Search of (15)	Nelder–Mead in $(\ln \kappa, \operatorname{logit} \pi_{\mathrm{b}}, \ln \mu_{\mathrm{rel}}, \ln \tau_{\mathrm{rel}})$ from a simplex of step 0.5, a wall of 10^6 times the squared distance outside Θ, stop at 10^{-2} in coordinates and objective or after 3000 evaluations, then L-BFGS-B with central differences of step 10^{-4}, at most 500 iterations, tolerances 10^{-12} in objective and 10^{-4} in gradient, 20 line search steps	the simplex reaches a basin without derivatives, and the polish refines its end
Restarts of (15)	from the best end, until one gains less than 0.01 nats, at most five	stops once a fresh search gains almost nothing
Training blocks	TWR-CLOUD 600 agent pairs and 600 anchor links per reading count from 40 scenarios of each training scan, TIERS, Bregar, HYMN, and AAU at most 2000 per reading count, IDLab at most four windows of ten consecutive readings per link setting	bound the blocks of a fit and balance them over reading counts, link types, and link settings
Multiplier fit (S8)	L-BFGS-B from $\alpha_\pm = 1$ with $\ln \alpha_\pm \in [-4, 4]$, tolerances 10^{-12} in objective and 10^{-8} in gradient	a convex problem, whose bounds keep a multiplier finite when the residuals of one sign are missing
Multipliers of TWR-CLOUD	per reading count, medians of 16 draws of the readings of every link in each training scan, the two kinds of Section S-I-G with equal weight	average over the draw of the readings
Multipliers of TIERS and IDLab	single readings at their true distances, every TIERS distance and orientation and every IDLab link setting with equal total weight, AF residuals divided by the constant reported scale, AAC residuals by $\min\{\max\{7\,\mathrm{cm} + 0.026\,d^\circ,\ 8\,\mathrm{cm}\},\ 55\,\mathrm{cm}\}$	balance distances, orientations, and link settings of unequal size (Section S-I-G)
Scale law of TWR-CLOUD	clipped to $[8, 55]$ cm	bounds the scale of very short and very long readings
Prior heights (S9)†	five from the 5th to the 95th percentile, one below a 10 cm span	cover the central 90 percent of the training heights, one prior height where they barely vary
Base budget of ANC†	step $\tilde{s}/4$ within $[5, 40]$ cm, 512 joint samples	a numerical budget from the reported anchor scale, bounded for cost and cell size (Section S-I-G)
Area padding†	largest anchor reading plus three times the largest reported anchor scale	a reading falls more than three scales short of its slant distance only after a gross error

†A length, a tolerance per centimeter, or a multiple of the reported scale, which another technology may set again (Section S-I-H).

of Section S-III can be read against their data. Outside TWR-CLOUD every method receives the same starting coordinates (Section S-V-B), drawn uniformly inside the operating area that both reading counts of a scenario share, and AF scenarios, which have no operating area, keep the starting coordinates built with them.

A. TWR-CLOUD

Eleven of the twelve scans of TWR-CLOUD [1], [2] place its 40 Qorvo DWM3001C radios at 130 of the 170 surveyed positions, and the twelfth, the evaluation scan, places them at the other 40. The release defines the scenarios of the twelfth scan. The 20 positions whose smallest pairwise distance is largest among the sets that hold ten required positions and none of four excluded ones form the anchor superset, from which it takes groups of 4, 8, 12, 16, and 20 anchors, and the other 20 positions hold the agents. For each count of 4, 7, 10, 13, 16, and 19 agents the release keeps at most 5000 agent groups, 24 710 in all. It poses every group without anchors for AF and with every anchor group for AAC and ANC, so AF holds 24 710 of the 148 260 scenarios, and AAC and ANC each hold the other 123 550. Every scenario has two iterations with readings of their own. Each direction of a link draws one reading and then up to ten readings from the reports of the scan, for the two reading counts, and an anchor link carries the one direction in which the agent answered the anchor. With both iterations, each reading count thus holds 49 420 solves in AF and 247 100 in AAC and in ANC, where a solve is one method positioning one scenario once. We draw scenarios from each training scan by the same rules for the fits of Section VI.

The adapter converts a recorded time of flight T into the range reading $y = (28.513\,\mathrm{cm/ns})\,T + 19.579\,\mathrm{cm}$ and reports the scale law $s(y)$ of Section S-I-A. The range line regresses the true slant distance on the time of flight over the reports of the training scans by iteratively reweighted least squares. Each report has weight one when its residual lies within 1.4826 median absolute deviations of the median residual, and otherwise that bound divided by the distance between the two. The scale law is the unweighted least-squares line through 1.4826 times the median absolute deviation of these residuals in 100 bins of equal count, ordered by true distance, against the median true distance of each bin. The law is thus fitted at the true distance and applied at the reading, between fixed bounds. Every method receives the starting coordinates of the release, the origin in the first iteration and random positions in the second, which SOLID uses only for an agent that its own start layouts cannot place.

The reporting cells of the release pair an anchor count with an agent count, 6 cells in AF and 30 in AAC and ANC, so every ANC scenario of TWR-CLOUD has four or more anchors and the main comparison uses all of them. The bootstrap draws agent groups within their agent count, since the release repeats every agent group under every anchor count, and the two iterations draw their readings from the same pool of scan reports. Its intervals are therefore conditional on the twelfth scan, its positions, and its readings. The bootstrap also treats the agent groups of an agent count as independent, although they share agents, and it inherits the reuse of reading seeds of the release.

B. TIERS

TIERS [3] records eight static Qorvo DWM1001 radios that range all 28 pairs in three recordings of about 30 000 reports each, within a published extent of 8 m by 9 m. Of the two report streams of a pair we keep the one from the lower to the higher identifier, since the reverse copies cannot be shown to be independent. A separate recording of two radios at 16 distances and 12 orientations, 19 199 reports against a motion-capture reference, calibrates the readings. Every method receives the same affine correction fitted on it, which multiplies a raw report by 0.98585, subtracts 21.09 cm, and clips the result at zero, with the constant scale of 11.17 cm. The 139 windows of ten reports common to all links of a recording form the scenarios, and with one reading a link uses only the first report of its window. Each window is one AF scenario of eight agents, and each subset of one to six of the eight radios taken as anchors makes one AAC and one ANC scenario, 246 per window.

In each recording the first half of the windows by rank trains, 22, 22, and 25 windows, and the rest evaluates, 22, 23, and 25 windows. Each window holds the next ten reports of every link, so the two parts split the reports of every link by their order and overlap in time across links that report at different rates. Both parts share the positions, the radios, and the shifts of the links. The evaluation part holds 70 AF scenarios and 17 220 scenarios each in AAC and ANC. Its reporting cells are the recordings in AF and the recordings with their anchor counts in AAC and ANC, and the bootstrap draws runs of three or four consecutive windows within each recording, so its intervals are conditional on the three recordings.

C. Bregar

The Bregar dataset [4], [5] places one tag at 85, 80, 84, and 81 surveyed points in four indoor environments, each with eight anchors, all DecaWave DWM1000 radios, and records 31 reports per point, anchor, and radio channel on six channels. The surveyed coordinates of these points replace inconsistent tag labels of the raw files, as references only, and the publisher's exclusion of anchor A6 in environment 2 holds for every method. Reports 1 to 10 of every link, the first window, form the training part, and reports 11 to 30, two windows, the evaluation part. All surveyed points of an environment and channel in one window form one ANC scenario, so the evaluation part holds 48 scenarios, balanced over the 24 reporting cells of environment and channel, and with one reading a link uses only the first report of the window. Such a scenario combines positions that one tag held at different times. ANC allows this, since its agents do not range each other and every agent keeps its own readings.

The adapter corrects each anchor and channel by a calibration constant, the mean of the first ten reports of its shortest line-of-sight link less its surveyed distance, and gives each

channel one root-mean-square scale over the corrected first windows of all links. The training part is thus positioned with constants taken from its own reports, while the evaluation part is corrected with the constants of the training part. These constants rest on one link in 80 to 85 per anchor, about 1.2 percent of the links, and the procedure of the publisher averages all 31 reports of that link instead. Every method leaves out a reading whose corrected distance is below zero, and no reading is clipped or refilled. The bootstrap draws the 24 reporting cells without strata, so its intervals are conditional on the four environments and conservative for them.

D. IDLab

The IDLab dataset [6], [7] records a fixed office network of 15 Wi-PoS radios with the Qorvo DW1000 chip, within a published extent of 41 m by 26 m, that range each other under many radio configurations of the physical layer. The published files hold 57 of the 72 configurations of its measurement schedule, 847 464 recorded reports, and we use them as published apart from five negative reports. The publisher's file office_lab_allsettings2.csv, in revision c1b2298 of the GitHub repository that the IDLab paper [6] names, gives the rows that each link and configuration occupies in these files. The first half of the recorded reports of each direction of every link and configuration forms the training part, 423 294 valid reports, and the rest the evaluation part, 424 165. Every method receives one affine correction of the reports with the constant scale of 80.26 cm, both fitted on the training part. For each configuration we put the evaluation reports of every link in a fixed hash order and cut them into windows of ten. The sweeps of a configuration take the windows of every link in turn, and a link with fewer windows reuses its windows, so each sweep poses the network with one to ten real readings per link. With one reading a sweep uses one report of each window, selected by a fixed hash. The 2708 sweeps are the AF scenarios. For AAC and ANC every sweep takes each count of 1 to 13 anchors in fifteen cyclic assignments of a hash-ordered list of the radios, which keeps at least two agents and makes every radio an anchor equally often, 528 060 scenarios in each of these modes. The training part forms its sweeps and scenarios from its own reports in the same way.

The reporting cells pair a configuration with an anchor count, and the bootstrap draws the configurations, since the sweeps of one configuration reuse its readings. The office network does not range every pair under every configuration, so the joined pairs cover 65.4 percent of the agent pairs in AF, 67.6 percent in AAC, and 40.9 percent in the main comparison of ANC, and a scenario without a joined pair is not scored.

E. HYMN

HYMN [8], [9] records one tag of a ZigPos real-time location system at 48 surveyed points, 42 in a hall and 6 outside it at its transitions, each for 127 to 145 epochs. An epoch is a round in which the tag ranges ten anchors at local heights of 3.1 to 4.2 m, against about 2 to 3.3 m for the points. A blank entry is a failed exchange of an anchor that the tag polled, since every epoch of the raw logs lists all ten anchors. The publisher documents no range calibration, so the readings enter as published, and one scale serves every reading, the root mean square of the admissible training residuals with equal weight per point and anchor, 220.65 cm. The 48 points, in the order of columns and rows of the publisher with the transition points last, alternate between the training and the evaluation part, so both parts span the hall.

A scenario takes eight points of one part as its agents, between one and ten of the ten anchors, and one of the first twelve windows of ten consecutive epochs of every agent, all drawn by a fixed hash, with sixty scenarios for each anchor count. Like a Bregar scenario, it combines positions that one tag held at different times. With ten readings a link uses the admissible readings in the window, and with one reading only the first epoch. If that epoch holds no admissible reading, the link contributes none, even when a later epoch holds one. The reporting cells are the anchor counts. The bootstrap draws the twelve windows, and its intervals are conditional on the 24 evaluation points and the ten anchors.

F. AAU Meshed UWB

AAU Meshed UWB [10], [11] records nine static Qorvo MDEK1001 radios, 0.33 to 3.26 m high within a published extent of 5 m by 8 m, that range each other in a fully meshed cycle. We use the exchanges among these nine radios in the fourteen recordings of measured readings and leave out the recordings of synthetic readings. The first seven recordings in time train and the last seven evaluate, and both parts hold the same nine positions and pairs. Every method receives the readings less the calibration constant of each ordered pair, which the publisher estimated on the first training recording, and one scale, the root mean square of the admissible calibrated training residuals with equal weight per ordered pair, 3.68 cm. No recording shows a relocated radio, a move of the median reading of a pair by more than 30 cm from the first recording.

In each recording the reports of every ordered pair, in time order, form windows of ten. For every window the nine radios are put in a fixed hash order, and for each anchor count from 1 to 7 the nine cyclic groups of that many radios are the anchors of nine scenarios, whose other radios are the agents. An anchor link carries the reports of the exchanges that the agent initiated toward the anchor, and with one reading it uses only the first report. The evaluation part thus holds 747 scenarios in each of its reporting cells, the anchor counts. The bootstrap draws runs of two or three consecutive windows within each recording.

The radios of HYMN and AAU report some exchanges with values that no ranging can yield, outside 0 to 59.96 m, the range of 200 ns of flight, HYMN up to 306 m and AAU mostly near 338 km of either sign. Among the training readings these are 128 of the 31 215 of HYMN and 5242 of the 76 612 of AAU. Their adapters keep these values out of every likelihood as inadmissible readings but still report the exchange, which counts as a success (Section IV-B).

S-III. Full Results

This section gives every result of the evaluation in full, which Section VIII summarizes by the S75 of Table II, and

ends with an ablation of the block likelihood on the training parts. Figs. S1 to S4 show the balanced CDFs of the pair error (Section VII-C) on HYMN and AAU, TIERS, Bregar, and IDLab. Tables S-IV to S-XV hold, for every dataset and mode, the balanced S50, S75, and S90 of every method under the joined pairs, each with the 95 percent interval of its paired cluster bootstrap in brackets. Below each baseline stands SOLID minus that baseline with its paired interval, so a negative difference whose interval excludes zero is a significant lead of SOLID at that percentile, and a positive one a significant deficit. The rows of AAC hold all reporting cells, which the main comparison uses, and then the scenarios with one anchor and with two anchors alone. The rows of ANC hold the scenarios with three or more anchors, which the main comparison uses, and then those with one and with two anchors. Each group of rows closes with the shares of the agent pairs and of the scenarios that the joined pairs score, which follow from the inputs alone and are the same for every method.

The Readings column gives the nominal reading count of Section III. The ranged pairs are those that a chain of links joins (Section IV-B). They give the same values as the joined pairs on every dataset except with one reading on HYMN with one anchor and on AAU, where some pairs succeeded only with readings that the adapter rejects (Section S-II-F), and Table S-XVI gives these rows. On every dataset the tables and curves compute S50, S75, and S90 as the balanced percentiles of the released scorer of TWR-CLOUD, and on TWR-CLOUD the scorer itself returns the same values, since it scores the joined pairs (Section VII-C).

Fig. S1 shows HYMN and AAU in ANC, with one reading per link in the upper row and ten in the lower row. On HYMN, SOLID lowers the S75 from 79.1 cm for ADE-BP to 54.4 cm with one reading and from 36.8 to 29.2 cm with ten. ADE-BP lies up to 2.2 cm left of SOLID below the median, and with one reading every baseline lies left of it near the 95th percentile. On AAU, SOLID lowers the S75 from 33.1 cm for FNL to 12.0 cm with one reading and from 9.2 cm for Jin et al. to 7.0 cm with ten. FNL lies left of SOLID below about the 57th percentile with one reading and the 68th with ten, with a significantly lower S50 of 4.4 against 5.2 cm with one reading and 2.7 against 3.9 cm with ten (Table S-VIII), but SOLID has a shorter tail than every baseline.

Fig. S2 shows TIERS, with one reading per link in the upper row and ten in the lower row, and AF, AAC, and ANC from left to right. SOLID lies left of every baseline in AF, and in AAC above about the 4th percentile, below which ADE-BP and, with ten readings, FNL lie left of it by at most 0.11 cm. In AAC, ADE-BP comes closest. The long tail of Jin et al. in AAC comes from the scenarios with one and two anchors, where their S50 with one reading is 373.3 and 113.9 cm, against 10.6 cm over all reporting cells (Table S-X), for the reason Section S-V-A gives. In ANC, ADE-BP lies slightly left of SOLID from about the 36th to the 95th percentile, and with ten readings also from about the 9th to the 19th. Its lead comes mostly from the recording whose eight radios stand on one line, each within 4.0 cm of their least-squares line. There the balanced S75 of SOLID is 10.94 and 10.21 cm with one and ten readings, against 9.65 and 8.92 cm for ADE-BP, while over the other two recordings it is 14.31 and 14.00 cm against 14.07 and 13.87 cm. Anchors on a line cannot tell a position from its mirror image across it, and with three to five anchors SOLID placed the agents a median of 17.4 to 21.9 cm off the line, against 2.2 to 2.8 cm for ADE-BP. With six anchors this median falls to 5.9 and 8.8 cm with one and ten readings.

Fig. S3 shows Bregar, with nominally one reading per link in the upper panel and ten in the lower one. ADE-BP lies left of SOLID from about the 29th percentile with one reading and from the 25th to the 96th percentile with ten, and its S50 and S75 lie significantly below those of SOLID (Table S-XII). Below these levels SOLID lies left of ADE-BP, apart from the 2nd to the 8th percentile with one reading. Jin et al. lie left of SOLID only between about the 33rd and the 54th percentile, STRONG only in short stretches below the 46th, and FNL lies about 10 cm or more right of SOLID at every level below the median.

Fig. S4 shows IDLab, where every method errs by meters on this sparsely linked office network. In AF Jin et al. lie left of SOLID between about the 12th and the 40th percentile and FNL between about the 12th and the 32nd, and SOLID lies left of both above the 40th. In AAC some baselines lie left of SOLID below about the 16th percentile, SOLID leads at the median, and FNL, ADE-BP, and STRONG lie left of it above about the 70th, 73rd, and 77th percentiles. FNL and ADE-BP thus have the lower S75, 388.9 and 403.4 cm against 413.9 cm with one reading and 387.0 and 402.6 cm against 413.7 cm with ten, by differences whose intervals include zero (Table S-XIV). With two anchors in AAC, FNL also has the lower S75, 410.4 and 408.6 cm against 422.7 and 419.0 cm with one and ten readings, by differences whose intervals include zero. In ANC SOLID lies left of the others from about the 20th to the 80th percentile, above which ADE-BP lies left of it, as the other baselines do above about the 90th.

Section VIII counts the 16 combinations of dataset, anchor count, and reading count that the ANC scenarios with one or two anchors form, and Tables S-VII, S-VIII, S-XI, and S-XV give the S75 of every method in each. In 12 of them FNL or STRONG has an S75 significantly below that of SOLID, on TIERS and HYMN with one and with two anchors, on IDLab with two anchors, and on AAU with ten readings. On HYMN with two anchors ADE-BP also lies below SOLID, by differences whose intervals include zero. SOLID lies significantly below the best baseline in the other four, on AAU with one reading and on IDLab with one anchor.

With one anchor the start of FNL and STRONG places every agent with a link on one ray from the anchor, and both stayed at it in every scenario on TIERS and AAU, STRONG also in 119 of its 120 solves on HYMN (Table S-XIX). With two anchors it places every agent linked to both anchors on one side of the line through them, or on that line where its two circles do not meet (Section S-I-D). On TIERS, AAU, and HYMN, 62.6, 64.6, and 70.5 percent of the joined pairs whose two agents lie more than 1 cm off that line have both on one side of it. FNL and STRONG both gave the true relation, one side or opposite sides, to 63.8 percent of the pairs they placed off the line on TIERS and to 62.7 percent on AAU, and on

HYMN, FNL to 68.5 and STRONG to 68.1 percent. SOLID, whose posteriors are symmetric about the line in their link part (Section V-B), gave it to 49.4, 48.6, and 51.7 percent on the three datasets. On IDLab only 53.1 percent of such pairs have both agents on one side. There the lead of STRONG comes from the 32.5 percent of joined pairs whose agents are linked to different anchors, since with the errors of STRONG on these pairs SOLID would have an S75 of 592.0 and 594.3 cm with one and ten readings. FNL errs on these pairs as STRONG does, by a mean of 711.3 and 710.7 cm, against 887.0 and 885.2 cm for SOLID. SOLID placed such a pair a mean of 479.4 and 476.3 cm farther apart than its two anchors, while the true pairs lie a mean of 11.3 cm closer together than their anchors, and STRONG placed them 7.8 and 10.6 cm closer.

Fig. S5 shows the time of one solve of every method in the controlled timing on TWR-CLOUD, from the median, as a dot, to the 95th percentile, as the end of the bar, on a logarithmic axis. That timing solved the whole evaluation scan with every method and mode, one run after another, in 90 single-threaded processes on 90 of the 96 physical cores of a machine with two Intel Xeon Platinum 8468 processors. Each run started on an idle machine, kept its evaluated budget, and returned the same estimates as the evaluation. Where FNL and STRONG start from the initializer layout of SOLID (Section S-V-B), their times include the time of building it. The 95th percentile of SOLID reaches 670, 1799, and 5520 ms in AF, AAC, and ANC, and in AAC that of ADE-BP is longer, 2691 ms.

The training parts also show which parts of the block likelihood carry the accuracy of SOLID. Table S-XVII gives the S75 of SOLID on the training parts of TWR-CLOUD, TIERS, Bregar, and IDLab, then its S75 with a Gaussian in place of the Student t distribution of (2), without the excess of (3), and without the offset, each time with the link parameters fitted again. The last column gives the S75 on the same scenarios of the baseline that Table II finds best in that row, with the options that the rule of Section VII-B selects and, for ADE-BP, the error law it fits on the training parts. Every scenario is positioned with link parameters that its own readings did not enter, fitted on the other ten training scans of TWR-CLOUD and elsewhere on the other of two folds of the training part, which split the windows of each TIERS recording, the channels of each Bregar environment, and the radio configurations of IDLab. On IDLab the table holds every scenario of one in eight of its training sweeps, chosen by a fixed hash. The standard error in parentheses is the standard deviation of the difference from SOLID over the 2000 draws of the paired cluster bootstrap of Section VII-C, with the clusters of Section S-II on TIERS, Bregar, and IDLab and the eleven training scans on TWR-CLOUD.

The heavy tail carries the lead of SOLID on TWR-CLOUD. With a Gaussian, its S75 there rises by 101 to 141 cm, above the best baseline of Table II in every combination but AF with one reading, while on TIERS and Bregar it changes by less than 1 cm and on IDLab by at most 4.3 cm. Dropping the excess raises the S75 on all four datasets. It rises by 4.9 to 9.3 cm on TWR-CLOUD, by 2.9 to 6.3 cm on TIERS, where SOLID would then have a higher S75 than the best baseline in every mode, by 36 and 37 cm on Bregar, and by 1.7 to

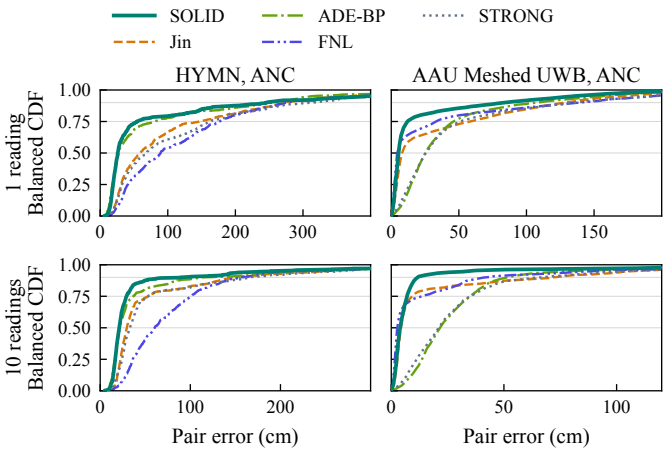


Fig. S1. Balanced CDFs of the pair error in ANC on HYMN and AAU.

6.7 cm on IDLab, where each of these differences lies within about two of its standard errors. The offset matters with ten readings in five of the six combinations of TWR-CLOUD and TIERS. Its removal then takes 9.2 of the 10.2 cm by which SOLID leads in AAC on TWR-CLOUD, raises the S75 by 7.5 cm in ANC there, and raises it by 1.9, 2.0, and 0.8 cm in AF, AAC, and ANC on TIERS. With one reading its removal lowers the S75 by 1.5 cm in ANC on TWR-CLOUD and by 3.5 cm on Bregar and changes it by less than 1 cm in the other combinations of TWR-CLOUD and TIERS, and on IDLab each of its changes lies within one standard error.

ADE-BP has the lower S75 in ANC on TIERS and on Bregar, and both leads are significant, as on the evaluation parts. It leads by 0.8 cm on TIERS with one and with ten readings, with paired intervals of 0.7 to 1.0 and 0.7 to 0.9 cm, and by 3.9 and 3.4 cm on Bregar with one and ten readings, with intervals of 1.2 to 5.6 and 1.3 to 5.2 cm. On TIERS the lead is largest in the recording whose radios stand on one line, where SOLID reaches 10.9 and 10.4 cm against 9.6 and 9.2 cm for ADE-BP with one and ten readings, while in the other two recordings it reaches 14.5 and 14.3 cm against 14.1 and 13.9 cm. On Bregar, where errors grow with the length of the link, the error law of ADE-BP widens with distance in three of the four environments, while the link model of SOLID gives short and long links the same law (Section IX). FNL has the lower S75 in AAC on IDLab, by 16.0 and 18.2 cm, differences whose paired intervals of -12.5 to 56.4 and -11.0 to 55.7 cm include zero, as on the evaluation part. In ANC on IDLab ADE-BP ran with its model 2 (Section S-V-A) on all of these scenarios and with the model 1 of Table S-XVIII on only some of them, so Table S-XVII gives it no value.

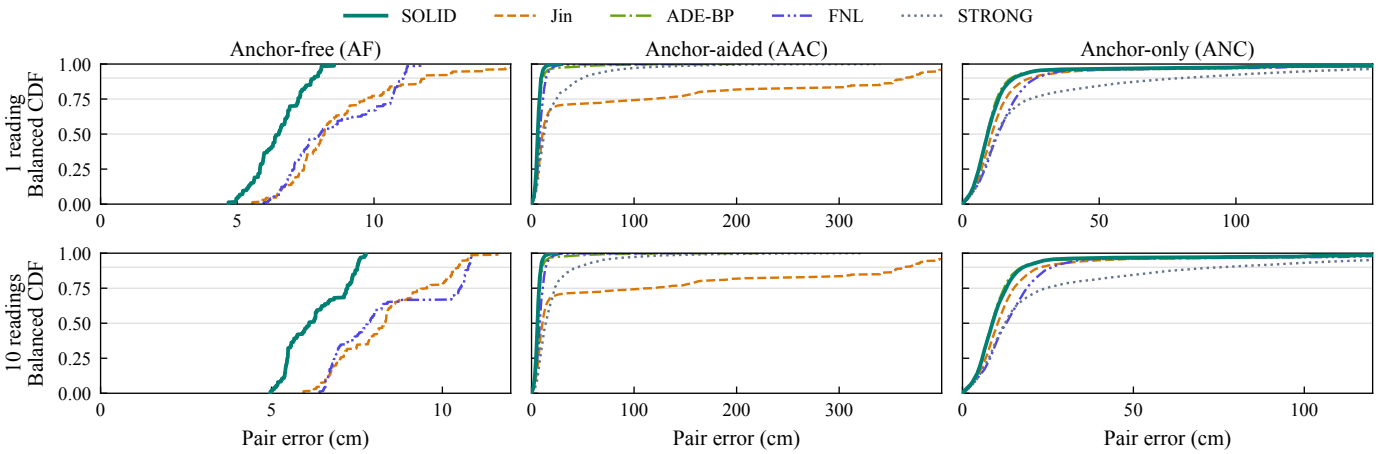


Fig. S2. Balanced CDFs of the pair error on TIERS.

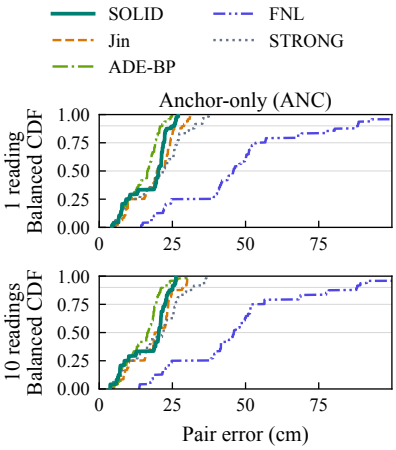


Fig. S3. Balanced CDFs of the pair error in ANC on Bregar.

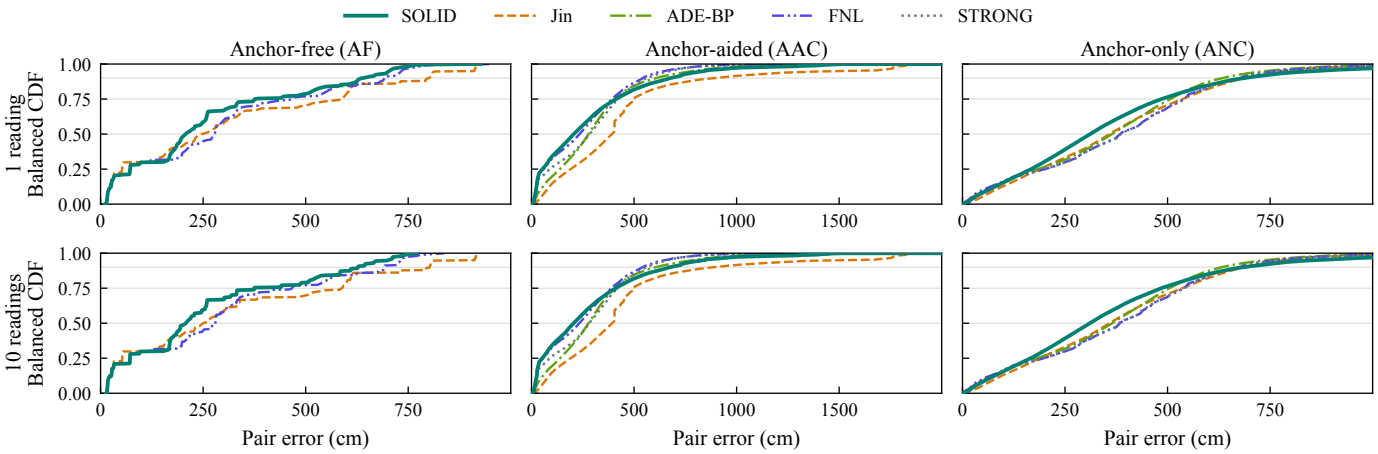


Fig. S4. Balanced CDFs of the pair error on IDLab.

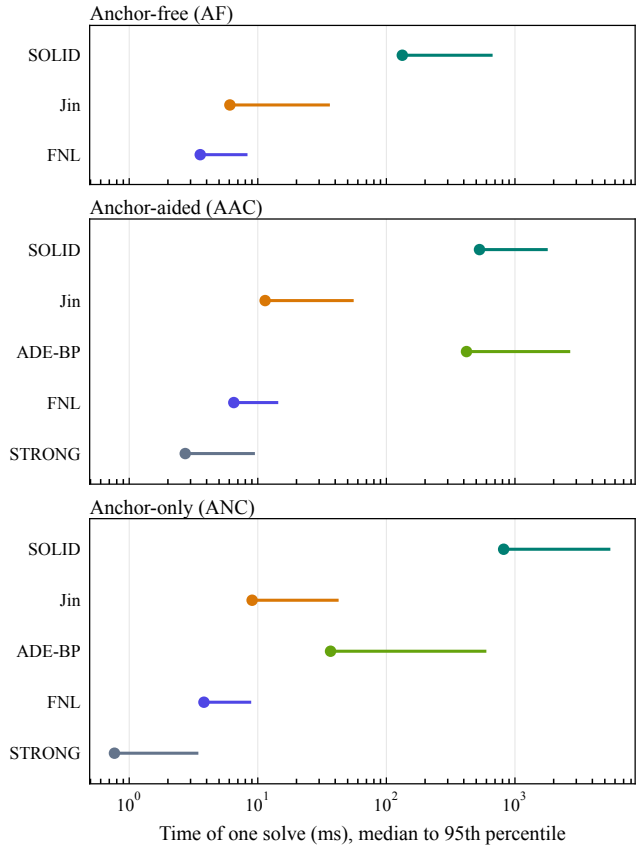


Fig. S5. Time of one solve on TWR-CLOUD.

TABLE S-IV
FULL RESULTS OF TWR-CLOUD IN AF

Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
1	SOLID	42.6 [41.1, 44.1]	125.0 [122.2, 127.8]	255.0 [251.0, 258.2]
	Jin	280.9 [272.7, 287.5]	372.4 [358.9, 386.7]	461.0 [448.2, 474.4]
	SOLID − Jin	−238.3 [−245.1, −230.3]	−247.4 [−261.4, −233.9]	−206.0 [−220.1, −193.0]
	FNL	210.6 [204.5, 218.6]	298.7 [286.7, 316.3]	392.9 [379.0, 404.0]
	SOLID − FNL	−168.0 [−176.2, −161.4]	−173.7 [−190.7, −162.2]	−137.9 [−150.1, −124.1]
	97.5 % of pairs and 100.0 % of scenarios scored			
10	SOLID	35.2 [33.0, 38.9]	98.4 [96.2, 100.6]	233.1 [228.6, 237.7]
	Jin	86.5 [80.4, 94.7]	213.5 [210.3, 216.5]	328.9 [325.7, 332.8]
	SOLID − Jin	−51.3 [−59.5, −45.5]	−115.1 [−118.2, −112.0]	−95.8 [−100.9, −91.0]
	FNL	40.1 [38.1, 42.1]	93.1 [90.6, 95.5]	244.5 [239.7, 248.9]
	SOLID − FNL	−4.9 [−6.5, −3.2]	+5.3 [+4.5, +6.3]	−11.4 [−13.1, −9.6]
	97.5 % of pairs and 100.0 % of scenarios scored			

TABLE S-V
FULL RESULTS OF TWR-CLOUD IN AAC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
All reporting cells	1	SOLID	14.7 [14.6, 14.8]	20.5 [20.3, 20.8]	48.6 [47.5, 50.1]
		Jin	41.6 [40.9, 42.2]	81.6 [80.2, 82.8]	142.6 [140.2, 145.5]
		SOLID − Jin	−26.9 [−27.5, −26.1]	−61.0 [−62.5, −59.6]	−94.1 [−97.4, −90.6]
		ADE-BP	23.9 [23.8, 24.2]	44.6 [43.9, 45.1]	101.2 [98.6, 103.1]
		SOLID − ADE-BP	−9.2 [−9.4, −9.1]	−24.1 [−24.7, −23.4]	−52.6 [−54.4, −50.3]
		FNL	128.1 [126.3, 130.0]	173.7 [171.2, 176.9]	223.7 [219.7, 228.3]
		SOLID − FNL	−113.4 [−115.3, −111.6]	−153.2 [−156.4, −150.7]	−175.1 [−180.1, −170.6]
		STRONG	117.7 [116.4, 119.0]	163.9 [161.4, 166.1]	205.9 [203.1, 209.3]
		SOLID − STRONG	−103.0 [−104.4, −101.6]	−143.4 [−145.6, −141.0]	−157.3 [−161.1, −153.8]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	13.9 [13.7, 14.0]	17.6 [17.5, 17.6]	30.6 [30.3, 31.0]
		Jin[a]	35.7 [35.5, 35.9]	44.8 [44.5, 45.2]	66.1 [65.1, 67.1]
		SOLID − Jin	−21.9 [−22.2, −21.6]	−27.3 [−27.6, −26.9]	−35.6 [−36.7, −34.5]
		ADE-BP	22.7 [22.5, 23.0]	37.9 [37.2, 38.6]	80.7 [77.9, 81.7]
		SOLID − ADE-BP	−8.9 [−9.2, −8.6]	−20.4 [−21.1, −19.6]	−50.1 [−51.2, −47.3]
		FNL	28.3 [27.8, 28.8]	40.6 [40.0, 41.2]	68.9 [67.8, 70.5]
		SOLID − FNL	−14.5 [−15.0, −14.0]	−23.1 [−23.7, −22.4]	−38.4 [−39.8, −37.3]
		STRONG	46.4 [46.1, 46.7]	70.3 [69.8, 70.7]	104.3 [103.8, 104.8]
		SOLID − STRONG	−32.6 [−32.9, −32.2]	−52.8 [−53.2, −52.2]	−73.7 [−74.3, −73.2]
		100.0 % of pairs and 100.0 % of scenarios scored			

[a]Failed on 1 solve.

TABLE S-VI
FULL RESULTS OF TWR-CLOUD IN ANC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
3 or more anchors	1	SOLID	27.4 [27.0, 27.8]	81.8 [79.7, 84.9]	199.1 [196.9, 202.2]
		Jin	59.0 [58.6, 59.6]	125.4 [123.0, 126.9]	235.1 [231.2, 238.3]
		SOLID − Jin	−31.6 [−32.3, −31.0]	−43.6 [−45.6, −40.6]	−36.0 [−39.8, −31.4]
		ADE-BP	52.5 [51.8, 53.0]	114.3 [112.6, 116.1]	212.6 [209.9, 216.8]
		SOLID − ADE-BP	−25.0 [−25.7, −24.2]	−32.5 [−34.0, −30.3]	−13.5 [−16.6, −11.2]
		FNL	118.0 [116.7, 119.7]	175.7 [174.3, 177.3]	227.0 [224.9, 228.6]
		SOLID − FNL	−90.6 [−92.5, −89.1]	−93.9 [−95.9, −91.2]	−27.9 [−29.7, −25.2]
		STRONG	67.2 [66.4, 68.3]	122.5 [120.4, 124.2]	288.6 [284.2, 291.6]
		SOLID − STRONG	−39.8 [−40.9, −38.9]	−40.8 [−42.3, −38.4]	−89.5 [−93.5, −84.1]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	22.6 [22.3, 22.9]	52.5 [52.2, 52.7]	179.8 [177.1, 182.2]
		Jin	41.3 [41.0, 41.6]	56.8 [56.5, 57.3]	228.0 [225.0, 232.0]
		SOLID − Jin	−18.7 [−19.0, −18.4]	−4.4 [−4.8, −4.0]	−48.2 [−51.6, −45.3]
		ADE-BP	45.2 [44.7, 45.7]	84.3 [83.9, 84.6]	190.8 [189.2, 193.6]
		SOLID − ADE-BP	−22.6 [−23.1, −22.2]	−31.8 [−32.1, −31.5]	−11.0 [−13.0, −9.3]
		FNL	54.1 [53.7, 54.9]	82.6 [82.2, 83.0]	198.4 [196.7, 199.2]
		SOLID − FNL	−31.5 [−32.3, −31.1]	−30.2 [−30.5, −29.8]	−18.6 [−20.4, −16.4]
		STRONG	38.9 [38.7, 38.9]	61.0 [60.8, 61.2]	284.7 [281.4, 286.4]
		SOLID − STRONG	−16.3 [−16.6, −16.0]	−8.6 [−8.8, −8.4]	−104.9 [−106.8, −101.9]
		100.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-VII
FULL RESULTS OF HYMN IN ANC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
3 or more anchors	1	SOLID	25.9 [23.8, 28.2]	54.4 [43.5, 75.4]	246.0 [165.4, 315.0]
		Jin	58.6 [47.1, 69.7]	146.3 [115.0, 176.1]	274.7 [252.8, 312.9]
		SOLID − Jin	−32.7 [−42.4, −23.1]	−91.9 [−120.7, −57.6]	−28.8 [−90.0, +28.2]
		ADE-BP	27.7 [23.8, 30.8]	79.1 [62.4, 109.4]	248.5 [221.5, 281.6]
		SOLID − ADE-BP	−1.8 [−3.1, +1.1]	−24.7 [−47.8, −9.9]	−2.5 [−58.2, +46.0]
		FNL	89.5 [68.1, 122.3]	176.5 [148.7, 204.8]	273.5 [243.6, 310.7]
		SOLID − FNL	−63.6 [−96.0, −42.2]	−122.1 [−148.0, −89.1]	−27.6 [−96.0, +29.4]
		STRONG	68.9 [53.5, 85.6]	163.4 [136.5, 186.3]	309.1 [247.6, 354.3]
		SOLID − STRONG	−43.0 [−59.9, −27.7]	−109.0 [−135.1, −81.7]	−63.2 [−130.4, −3.9]
		99.9 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	20.9 [19.7, 21.9]	29.2 [27.3, 31.6]	89.3 [48.5, 143.7]
		Jin	29.5 [27.7, 31.2]	53.2 [42.9, 75.6]	149.7 [132.4, 179.0]
		SOLID − Jin	−8.6 [−11.1, −6.6]	−23.9 [−45.7, −14.4]	−60.4 [−114.1, −10.4]
		ADE-BP	21.7 [20.5, 23.1]	36.8 [31.0, 40.3]	111.7 [89.0, 173.4]
		SOLID − ADE-BP	−0.8 [−2.6, +0.6]	−7.6 [−11.9, −2.9]	−22.4 [−85.5, +19.1]
		FNL	60.5 [52.4, 65.7]	101.0 [90.6, 111.0]	156.0 [138.0, 179.2]
		SOLID − FNL	−39.7 [−45.2, −31.7]	−71.8 [−79.5, −62.4]	−66.7 [−99.9, −27.7]
		STRONG	33.0 [30.8, 35.7]	51.9 [45.5, 66.8]	166.9 [135.3, 202.4]
		SOLID − STRONG	−12.1 [−14.9, −9.6]	−22.7 [−37.3, −16.5]	−77.6 [−122.6, −33.0]
		100.0 % of pairs and 100.0 % of scenarios scored			
1 anchor	1	SOLID	1034.2 [908.3, 1145.1]	1273.1 [1185.7, 1427.7]	1758.0 [1421.1, 1837.4]
		Jin	1756.5 [1694.9, 1819.4]	1887.9 [1828.0, 1939.4]	1980.4 [1934.8, 2182.9]
		SOLID − Jin	−722.2 [−854.9, −564.1]	−614.8 [−707.5, −410.0]	−222.4 [−679.2, −115.7]
		ADE-BP	1323.8 [1237.7, 1370.9]	1478.5 [1396.8, 1545.6]	1609.5 [1534.9, 1819.0]
		SOLID − ADE-BP	−289.6 [−429.3, −129.3]	−205.4 [−318.9, −16.8]	+148.5 [−389.1, +273.5]
		FNL	487.6 [391.2, 683.7]	899.4 [683.7, 1022.8]	1065.7 [959.2, 1344.2]
		SOLID − FNL	+546.7 [+263.1, +715.4]	+373.8 [+202.9, +737.4]	+692.3 [+208.2, +823.8]
		STRONG	487.6 [391.2, 683.7]	899.4 [683.7, 1022.8]	1065.7 [959.2, 1344.2]
		SOLID − STRONG	+546.7 [+263.1, +715.4]	+373.8 [+202.9, +737.4]	+692.3 [+208.2, +823.8]
		87.6 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	990.9 [951.0, 1230.6]	1358.2 [1248.9, 1673.9]	1793.4 [1439.4, 1935.7]
		Jin	1773.0 [1727.8, 1842.1]	1936.2 [1842.1, 1981.8]	2065.9 [1953.4, 2182.9]
		SOLID − Jin	−782.0 [−877.7, −538.7]	−578.0 [−708.8, −264.9]	−272.6 [−647.1, −150.6]
		ADE-BP	1312.6 [1240.2, 1428.7]	1491.6 [1365.8, 1570.2]	1597.9 [1536.9, 1747.4]
		SOLID − ADE-BP	−321.6 [−460.8, −31.8]	−133.3 [−286.4, +172.7]	+195.5 [−216.4, +285.7]
		FNL	602.7 [453.8, 821.5]	957.3 [791.8, 1024.6]	1067.7 [965.3, 1208.9]
		SOLID − FNL	+388.2 [+158.9, +730.7]	+400.9 [+272.4, +822.9]	+725.7 [+317.4, +868.0]
		STRONG	599.5 [423.6, 795.8]	912.4 [718.9, 1022.4]	1059.7 [965.1, 1205.4]
		SOLID − STRONG	+391.5 [+176.1, +771.2]	+445.8 [+270.0, +834.8]	+733.7 [+320.8, +869.9]
		95.1 % of pairs and 100.0 % of scenarios scored			
2 anchors	1	SOLID	565.4 [478.1, 725.9]	856.0 [686.7, 1157.3]	1386.8 [969.1, 1671.7]
		Jin	591.9 [489.8, 740.6]	901.3 [769.6, 1055.3]	1139.7 [990.0, 1296.8]
		SOLID − Jin	−26.4 [−196.2, +143.6]	−45.2 [−226.0, +283.1]	+247.1 [−198.0, +510.7]
		ADE-BP	559.9 [503.4, 674.4]	855.4 [726.4, 1055.9]	1115.1 [1040.2, 1419.7]
		SOLID − ADE-BP	+5.6 [−115.2, +141.7]	+0.7 [−166.5, +224.3]	+271.7 [−155.6, +396.2]
		FNL	389.9 [282.6, 636.9]	651.5 [548.4, 926.6]	938.4 [689.5, 1208.5]
		SOLID − FNL	+175.6 [−14.2, +329.6]	+204.5 [+43.4, +465.7]	+448.4 [−23.0, +722.0]
		STRONG	389.8 [282.6, 633.4]	651.2 [548.4, 926.6]	938.1 [691.6, 1208.6]
		SOLID − STRONG	+175.6 [−8.6, +329.7]	+204.8 [+43.6, +465.7]	+448.7 [+70.6, +722.0]
		97.5 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	397.2 [303.2, 450.9]	644.3 [471.6, 847.0]	970.5 [742.8, 1628.4]
		Jin	445.0 [366.8, 529.3]	652.2 [561.5, 966.0]	979.3 [862.2, 1230.2]
		SOLID − Jin	−47.8 [−141.9, +27.8]	−7.9 [−184.7, +144.0]	−8.8 [−229.5, +610.5]
		ADE-BP	378.9 [336.7, 471.4]	574.8 [441.3, 860.4]	981.1 [784.8, 1591.2]
		SOLID − ADE-BP	+18.2 [−49.1, +75.8]	+69.5 [−121.1, +177.9]	−10.6 [−537.6, +190.6]
		FNL	232.2 [198.5, 283.6]	336.4 [255.8, 419.5]	562.1 [399.4, 816.9]
		SOLID − FNL	+165.0 [+58.4, +248.5]	+307.8 [+112.3, +508.1]	+408.5 [+93.8, +1116.1]
		STRONG	206.7 [143.4, 244.2]	301.1 [244.2, 396.8]	561.1 [368.4, 818.2]
		SOLID − STRONG	+190.4 [+83.5, +293.4]	+343.2 [+130.3, +541.5]	+409.4 [+154.8, +1122.8]
		99.2 % of pairs and 100.0 % of scenarios scored			

TABLE S-VIII
FULL RESULTS OF AAU MESHED UWB IN ANC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
3 or more anchors	1	SOLID	5.2 [4.9, 5.5]	12.0 [10.6, 15.4]	84.7 [69.3, 100.8]
		Jin	8.0 [7.3, 8.7]	58.2 [49.3, 68.6]	127.6 [118.8, 137.2]
		SOLID − Jin	−2.8 [−3.5, −2.2]	−46.2 [−53.6, −38.3]	−43.0 [−54.9, −32.3]
		ADE-BP	25.3 [24.2, 26.2]	45.9 [43.1, 49.7]	107.8 [91.6, 119.7]
		SOLID − ADE-BP	−20.1 [−21.1, −19.1]	−33.9 [−36.3, −31.6]	−23.2 [−32.6, −13.0]
		FNL	4.4 [4.1, 4.9]	33.1 [29.2, 39.3]	136.5 [120.4, 155.7]
		SOLID − FNL	+0.7 [+0.4, +1.0]	−21.1 [−24.5, −18.2]	−51.8 [−64.8, −42.5]
		STRONG	25.4 [24.3, 26.9]	50.8 [45.8, 59.7]	137.4 [120.1, 154.1]
		SOLID − STRONG	−20.2 [−21.7, −19.2]	−38.8 [−44.7, −34.7]	−52.7 [−63.8, −42.8]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	3.9 [3.8, 4.2]	7.0 [6.7, 7.2]	11.6 [11.0, 12.5]
		Jin	4.3 [4.1, 4.4]	9.2 [8.6, 9.6]	71.6 [70.0, 74.7]
		SOLID − Jin	−0.3 [−0.5, −0.1]	−2.3 [−2.6, −1.7]	−60.0 [−63.0, −58.1]
		ADE-BP	21.6 [21.2, 22.1]	35.0 [33.8, 36.2]	51.4 [48.6, 54.0]
		SOLID − ADE-BP	−17.7 [−18.1, −17.2]	−28.0 [−29.2, −26.8]	−39.8 [−42.1, −37.1]
		FNL	2.7 [2.5, 2.9]	13.6 [11.0, 16.6]	42.0 [38.2, 45.7]
		SOLID − FNL	+1.2 [+1.1, +1.4]	−6.6 [−9.7, −4.0]	−30.5 [−33.6, −26.8]
		STRONG	21.1 [20.2, 21.9]	35.0 [33.2, 36.4]	66.3 [59.7, 70.6]
		SOLID − STRONG	−17.2 [−17.9, −16.4]	−28.0 [−29.4, −26.3]	−54.7 [−58.7, −48.5]
		100.0 % of pairs and 100.0 % of scenarios scored			
1 anchor	1	SOLID	294.8 [293.1, 296.6]	312.4 [310.4, 314.6]	330.6 [327.3, 333.8]
		Jin	473.0 [473.0, 473.0]	486.7 [486.7, 486.7]	499.7 [499.7, 499.7]
		SOLID − Jin	−178.2 [−179.9, −176.4]	−174.3 [−176.2, −172.0]	−169.1 [−172.4, −165.9]
		ADE-BP	385.9 [382.7, 387.3]	412.6 [410.6, 413.8]	429.9 [427.2, 433.3]
		SOLID − ADE-BP	−91.1 [−93.1, −87.5]	−100.2 [−102.8, −96.4]	−99.2 [−103.0, −95.3]
		FNL	269.2 [268.1, 270.1]	328.9 [320.5, 331.5]	369.5 [359.8, 390.2]
		SOLID − FNL	+25.7 [+24.0, +27.7]	−16.6 [−19.3, −8.6]	−38.8 [−59.3, −29.3]
		STRONG	269.2 [268.1, 270.1]	328.9 [320.5, 331.5]	369.5 [359.8, 390.2]
		SOLID − STRONG	+25.7 [+24.0, +27.7]	−16.6 [−19.3, −8.6]	−38.8 [−59.3, −29.3]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	291.7 [289.7, 293.2]	305.7 [305.0, 306.7]	317.1 [314.8, 318.5]
		Jin	473.0 [473.0, 473.0]	486.7 [486.7, 486.7]	499.7 [499.7, 499.7]
		SOLID − Jin	−181.3 [−183.4, −179.9]	−180.9 [−181.7, −179.9]	−182.6 [−184.9, −181.2]
		ADE-BP	385.4 [381.8, 387.8]	412.1 [409.7, 414.0]	431.1 [428.1, 434.1]
		SOLID − ADE-BP	−93.7 [−97.3, −90.1]	−106.3 [−108.7, −103.5]	−114.0 [−118.1, −110.7]
		FNL	241.3 [241.1, 241.4]	303.5 [303.2, 303.6]	394.0 [393.8, 394.1]
		SOLID − FNL	+50.4 [+48.4, +51.9]	+2.2 [+1.4, +3.3]	−77.0 [−79.2, −75.5]
		STRONG	241.3 [241.1, 241.4]	303.5 [303.2, 303.6]	394.0 [393.8, 394.1]
		SOLID − STRONG	+50.4 [+48.4, +51.9]	+2.2 [+1.4, +3.3]	−77.0 [−79.2, −75.5]
		100.0 % of pairs and 100.0 % of scenarios scored			
2 anchors	1	SOLID	171.3 [165.3, 175.6]	206.4 [200.9, 212.7]	286.2 [262.3, 317.2]
		Jin	218.1 [213.0, 222.2]	280.6 [274.8, 284.8]	373.2 [360.0, 377.8]
		SOLID − Jin	−46.8 [−51.6, −42.7]	−74.3 [−79.4, −68.0]	−87.0 [−105.0, −54.1]
		ADE-BP	188.0 [183.3, 193.7]	227.5 [222.3, 233.4]	275.5 [265.5, 288.1]
		SOLID − ADE-BP	−16.7 [−22.8, −11.6]	−21.1 [−28.1, −14.8]	+10.8 [−13.1, +36.3]
		FNL	155.5 [144.7, 166.2]	218.2 [210.9, 221.6]	268.2 [255.8, 276.8]
		SOLID − FNL	+15.8 [+3.2, +28.2]	−11.8 [−17.5, −4.0]	+18.1 [−6.7, +53.7]
		STRONG	155.5 [144.7, 166.3]	218.2 [210.9, 221.6]	268.2 [255.3, 276.8]
		SOLID − STRONG	+15.8 [+3.0, +28.2]	−11.8 [−17.5, −4.0]	+18.1 [−6.7, +53.7]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	158.7 [150.1, 165.6]	195.7 [191.7, 200.6]	282.2 [235.3, 313.4]
		Jin	193.0 [191.5, 207.8]	272.1 [265.1, 273.6]	362.3 [360.9, 375.2]
		SOLID − Jin	−34.2 [−45.2, −29.6]	−76.4 [−79.8, −70.0]	−80.1 [−127.2, −48.0]
		ADE-BP	176.4 [170.7, 179.3]	208.7 [205.5, 213.6]	262.3 [244.3, 290.4]
		SOLID − ADE-BP	−17.7 [−25.2, −10.2]	−13.0 [−16.6, −9.0]	+19.9 [−19.1, +51.0]
		FNL	115.4 [113.3, 118.4]	145.2 [144.1, 166.8]	183.4 [180.5, 184.0]
		SOLID − FNL	+43.3 [+33.5, +50.4]	+50.5 [+29.5, +55.1]	+98.8 [+51.7, +130.9]
		STRONG	115.4 [113.3, 118.4]	145.2 [144.1, 166.8]	183.4 [177.8, 184.0]
		SOLID − STRONG	+43.3 [+33.5, +50.4]	+50.5 [+29.5, +55.1]	+98.8 [+51.7, +130.9]
		100.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-IX
FULL RESULTS OF TIERS IN AF

Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
1	SOLID	6.5 [6.3, 6.6]	7.2 [6.9, 7.5]	7.8 [7.4, 8.0]
	Jin	8.2 [7.8, 8.5]	9.7 [9.0, 10.6]	11.9 [10.6, 14.0]
	SOLID − Jin	−1.7 [−1.9, −1.4]	−2.5 [−3.4, −1.8]	−4.1 [−6.3, −2.9]
	FNL	8.0 [7.6, 8.6]	10.7 [10.1, 10.8]	11.1 [10.8, 11.2]
	SOLID − FNL	−1.5 [−2.0, −1.1]	−3.4 [−3.7, −2.8]	−3.3 [−3.5, −2.9]
	100.0 % of pairs and 100.0 % of scenarios scored			
10	SOLID	6.1 [6.0, 6.3]	7.2 [7.1, 7.4]	7.5 [7.3, 7.6]
	Jin	8.3 [8.0, 8.4]	9.6 [9.1, 10.1]	10.5 [10.2, 10.5]
	SOLID − Jin	−2.2 [−2.4, −1.7]	−2.3 [−2.9, −1.9]	−2.9 [−3.2, −2.7]
	FNL	7.8 [7.4, 8.0]	10.5 [10.4, 10.6]	10.8 [10.6, 10.8]
	SOLID − FNL	−1.7 [−1.9, −1.4]	−3.3 [−3.4, −3.2]	−3.2 [−3.4, −3.2]
	100.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-X
FULL RESULTS OF TIERS IN AAC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
All reporting cells	1	SOLID	6.0 [5.9, 6.1]	7.6 [7.4, 7.7]	9.3 [9.1, 9.5]
		Jin	10.6 [10.3, 10.9]	117.6 [116.3, 118.9]	362.4 [362.4, 362.4]
		SOLID − Jin	−4.6 [−4.8, −4.3]	−110.0 [−111.3, −108.6]	−353.1 [−353.2, −352.9]
		ADE-BP	6.4 [6.3, 6.5]	8.2 [8.1, 8.4]	11.0 [10.6, 11.4]
		SOLID − ADE-BP	−0.4 [−0.5, −0.3]	−0.6 [−0.7, −0.5]	−1.7 [−2.0, −1.4]
		FNL	8.2 [8.1, 8.3]	11.2 [11.0, 11.4]	14.1 [13.9, 14.3]
		SOLID − FNL	−2.2 [−2.3, −2.1]	−3.6 [−3.8, −3.4]	−4.8 [−5.0, −4.6]
		STRONG	12.1 [11.8, 12.5]	24.0 [23.5, 24.6]	49.0 [48.3, 50.9]
		SOLID − STRONG	−6.1 [−6.5, −5.8]	−16.4 [−17.0, −15.9]	−39.7 [−41.6, −38.9]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	5.8 [5.7, 5.8]	7.3 [7.2, 7.4]	9.0 [8.9, 9.1]
		Jin	10.5 [10.3, 10.6]	116.7 [116.0, 117.3]	362.4 [362.4, 362.4]
		SOLID − Jin	−4.7 [−4.8, −4.6]	−109.4 [−110.0, −108.6]	−353.4 [−353.5, −353.3]
		ADE-BP	6.2 [6.1, 6.3]	8.0 [7.9, 8.1]	10.7 [10.5, 10.9]
		SOLID − ADE-BP	−0.5 [−0.5, −0.4]	−0.7 [−0.8, −0.6]	−1.7 [−1.9, −1.5]
		FNL	8.0 [7.9, 8.1]	11.0 [10.9, 11.1]	13.8 [13.6, 13.9]
		SOLID − FNL	−2.3 [−2.4, −2.2]	−3.7 [−3.8, −3.6]	−4.8 [−4.9, −4.6]
		STRONG	13.7 [13.5, 13.9]	22.5 [22.1, 23.1]	45.4 [44.2, 46.4]
		SOLID − STRONG	−8.0 [−8.1, −7.8]	−15.2 [−15.8, −14.8]	−36.4 [−37.4, −35.1]
		100.0 % of pairs and 100.0 % of scenarios scored			
1 anchor	1	SOLID	6.4 [6.2, 6.5]	7.4 [7.3, 7.6]	8.2 [8.0, 8.5]
		Jin	373.3 [373.3, 373.3]	397.5 [397.5, 397.5]	438.6 [438.6, 438.6]
		SOLID − Jin	−366.9 [−367.1, −366.8]	−390.1 [−390.2, −389.9]	−430.4 [−430.7, −430.2]
		ADE-BP	7.1 [6.8, 7.2]	8.6 [8.3, 9.1]	14.4 [12.7, 22.5]
		SOLID − ADE-BP	−0.7 [−0.8, −0.5]	−1.2 [−1.6, −0.9]	−6.2 [−13.9, −4.5]
		FNL	8.1 [8.0, 8.2]	10.9 [10.5, 11.3]	12.0 [11.8, 12.1]
		SOLID − FNL	−1.7 [−1.9, −1.6]	−3.5 [−3.8, −3.0]	−3.8 [−4.0, −3.4]
		STRONG	23.6 [23.4, 24.1]	36.2 [35.4, 36.4]	45.5 [44.7, 46.1]
		SOLID − STRONG	−17.2 [−17.8, −16.9]	−28.8 [−29.1, −27.9]	−37.3 [−37.9, −36.4]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	6.0 [5.9, 6.2]	7.2 [7.1, 7.3]	8.0 [7.9, 8.1]
		Jin	373.3 [373.3, 373.3]	397.5 [397.5, 397.5]	438.6 [438.6, 438.6]
		SOLID − Jin	−367.3 [−367.4, −367.1]	−390.3 [−390.4, −390.2]	−430.6 [−430.7, −430.5]
		ADE-BP	6.9 [6.8, 7.2]	8.6 [8.3, 9.2]	13.7 [11.9, 17.1]
		SOLID − ADE-BP	−0.9 [−1.1, −0.7]	−1.4 [−2.0, −1.2]	−5.6 [−9.1, −3.9]
		FNL	7.9 [7.7, 8.1]	10.2 [9.5, 10.2]	11.7 [11.6, 11.8]
		SOLID − FNL	−1.9 [−2.0, −1.7]	−3.0 [−3.0, −2.3]	−3.7 [−3.8, −3.6]
		STRONG	20.5 [20.2, 21.0]	24.7 [24.2, 24.9]	31.1 [31.0, 31.4]
		SOLID − STRONG	−14.5 [−15.0, −14.2]	−17.5 [−17.7, −17.0]	−23.1 [−23.5, −22.9]
		100.0 % of pairs and 100.0 % of scenarios scored			
2 anchors	1	SOLID	6.3 [6.2, 6.4]	7.6 [7.4, 7.7]	8.6 [8.5, 8.9]
		Jin	113.9 [111.4, 113.9]	159.9 [159.3, 160.0]	195.9 [194.7, 196.2]
		SOLID − Jin	−107.6 [−107.7, −105.2]	−152.3 [−152.6, −151.7]	−187.2 [−187.7, −186.1]
		ADE-BP	6.7 [6.5, 6.8]	8.1 [8.0, 8.3]	9.7 [9.4, 10.1]
		SOLID − ADE-BP	−0.4 [−0.5, −0.3]	−0.6 [−0.7, −0.4]	−1.1 [−1.3, −0.8]
		FNL	8.1 [8.0, 8.3]	10.0 [9.7, 10.4]	12.6 [12.4, 12.9]
		SOLID − FNL	−1.9 [−2.0, −1.7]	−2.4 [−2.8, −2.2]	−4.0 [−4.2, −3.6]
		STRONG	15.3 [14.7, 16.0]	24.4 [24.1, 25.1]	43.1 [42.6, 43.5]
		SOLID − STRONG	−9.1 [−9.6, −8.5]	−16.8 [−17.5, −16.5]	−34.4 [−34.8, −33.9]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	6.0 [5.9, 6.1]	7.3 [7.2, 7.4]	8.4 [8.3, 8.6]
		Jin	109.8 [109.2, 109.8]	159.3 [159.1, 159.5]	195.1 [194.1, 195.2]
		SOLID − Jin	−103.7 [−103.8, −103.2]	−152.0 [−152.3, −151.7]	−186.6 [−186.8, −185.7]
		ADE-BP	6.4 [6.3, 6.6]	8.0 [7.8, 8.1]	9.7 [9.4, 10.1]
		SOLID − ADE-BP	−0.4 [−0.5, −0.3]	−0.6 [−0.9, −0.5]	−1.3 [−1.7, −1.0]
		FNL	8.0 [7.9, 8.1]	9.6 [9.4, 9.8]	12.5 [12.4, 12.6]
		SOLID − FNL	−2.0 [−2.0, −1.9]	−2.3 [−2.4, −2.2]	−4.1 [−4.2, −4.0]
		STRONG	18.8 [18.6, 19.0]	24.6 [24.5, 25.0]	39.7 [39.2, 39.8]
		SOLID − STRONG	−12.8 [−13.0, −12.6]	−17.3 [−17.6, −17.1]	−31.2 [−31.4, −30.8]
		100.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-XI
FULL RESULTS OF TIERS IN ANC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
3 or more anchors	1	SOLID	9.2 [9.0, 9.3]	13.1 [12.9, 13.4]	18.7 [18.4, 19.1]
		Jin	10.7 [10.4, 11.0]	15.8 [15.4, 16.1]	24.8 [23.8, 25.6]
		SOLID − Jin	−1.5 [−1.7, −1.3]	−2.7 [−2.9, −2.4]	−6.1 [−6.7, −5.4]
		ADE-BP	9.0 [8.8, 9.1]	12.5 [12.3, 12.7]	17.7 [17.4, 18.2]
		SOLID − ADE-BP	+0.2 [+0.1, +0.3]	+0.6 [+0.5, +0.8]	+0.9 [+0.7, +1.2]
		FNL	13.0 [12.8, 13.2]	19.2 [18.8, 19.6]	26.8 [26.3, 27.4]
		SOLID − FNL	−3.8 [−4.0, −3.6]	−6.1 [−6.4, −5.8]	−8.1 [−8.6, −7.7]
		STRONG	12.5 [12.1, 12.8]	24.4 [23.3, 26.0]	79.7 [76.7, 82.6]
		SOLID − STRONG	−3.3 [−3.5, −3.0]	−11.3 [−12.9, −10.2]	−61.0 [−63.9, −58.0]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	8.7 [8.7, 8.8]	12.8 [12.6, 12.9]	17.7 [17.4, 18.3]
		Jin	10.3 [10.2, 10.5]	15.0 [14.8, 15.2]	22.9 [22.5, 23.3]
		SOLID − Jin	−1.6 [−1.7, −1.5]	−2.2 [−2.4, −2.1]	−5.1 [−5.5, −4.7]
		ADE-BP	8.5 [8.4, 8.6]	12.0 [11.9, 12.2]	17.3 [17.0, 17.6]
		SOLID − ADE-BP	+0.2 [+0.1, +0.3]	+0.7 [+0.6, +0.8]	+0.5 [+0.2, +0.8]
		FNL	12.8 [12.6, 13.0]	18.8 [18.5, 19.1]	26.3 [25.9, 26.7]
		SOLID − FNL	−4.1 [−4.2, −3.9]	−6.0 [−6.2, −5.9]	−8.6 [−8.8, −8.2]
		STRONG	12.4 [12.2, 12.6]	24.4 [23.4, 25.4]	75.6 [72.3, 78.7]
		SOLID − STRONG	−3.7 [−3.9, −3.5]	−11.6 [−12.6, −10.7]	−57.9 [−61.0, −54.5]
		100.0 % of pairs and 100.0 % of scenarios scored			
1 anchor	1	SOLID	224.8 [222.9, 227.5]	272.8 [271.5, 274.4]	327.1 [325.6, 331.0]
		Jin	373.3 [373.3, 373.3]	397.5 [397.5, 397.5]	438.6 [438.6, 438.6]
		SOLID − Jin	−148.5 [−150.4, −145.8]	−124.7 [−126.0, −123.1]	−111.5 [−113.0, −107.6]
		ADE-BP	292.7 [286.6, 295.7]	325.8 [323.9, 329.9]	352.1 [347.4, 356.2]
		SOLID − ADE-BP	−68.0 [−71.4, −62.2]	−53.0 [−57.2, −50.2]	−24.9 [−30.1, −18.3]
		FNL	179.1 [178.7, 179.5]	225.1 [224.7, 225.4]	267.5 [267.1, 268.1]
		SOLID − FNL	+45.7 [+43.6, +48.3]	+47.7 [+46.4, +49.7]	+59.6 [+57.9, +63.5]
		STRONG	179.1 [178.7, 179.5]	225.1 [224.7, 225.4]	267.5 [267.1, 268.1]
		SOLID − STRONG	+45.7 [+43.6, +48.3]	+47.7 [+46.4, +49.7]	+59.6 [+57.9, +63.5]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	225.0 [223.3, 227.1]	273.6 [272.2, 274.7]	330.5 [325.9, 331.8]
		Jin	373.3 [373.3, 373.3]	397.5 [397.5, 397.5]	438.6 [438.6, 438.6]
		SOLID − Jin	−148.3 [−150.0, −146.2]	−123.9 [−125.3, −122.8]	−108.1 [−112.7, −106.8]
		ADE-BP	291.6 [289.6, 294.6]	326.3 [322.5, 327.7]	353.9 [347.0, 355.5]
		SOLID − ADE-BP	−66.6 [−70.1, −63.5]	−52.8 [−54.9, −48.6]	−23.4 [−28.5, −16.2]
		FNL	178.9 [178.1, 178.9]	224.8 [224.7, 224.9]	267.2 [266.9, 267.6]
		SOLID − FNL	+46.1 [+44.7, +48.7]	+48.8 [+47.4, +49.9]	+63.4 [+58.5, +64.7]
		STRONG	178.9 [178.1, 178.9]	224.8 [224.7, 224.9]	267.2 [266.9, 267.6]
		SOLID − STRONG	+46.1 [+44.7, +48.7]	+48.8 [+47.4, +49.9]	+63.4 [+58.5, +64.7]
		100.0 % of pairs and 100.0 % of scenarios scored			
2 anchors	1	SOLID	70.8 [69.8, 71.9]	119.5 [116.8, 121.3]	142.2 [141.1, 144.4]
		Jin	110.1 [108.7, 111.6]	152.0 [151.6, 153.1]	187.5 [187.2, 188.6]
		SOLID − Jin	−39.3 [−41.0, −37.5]	−32.5 [−35.8, −30.4]	−45.3 [−47.1, −43.1]
		ADE-BP	74.7 [72.4, 76.9]	122.7 [119.4, 124.9]	147.4 [144.5, 149.4]
		SOLID − ADE-BP	−3.9 [−6.0, −1.7]	−3.2 [−6.1, −0.5]	−5.2 [−7.0, −2.4]
		FNL	26.6 [26.1, 27.2]	94.6 [94.0, 95.1]	122.4 [119.9, 124.0]
		SOLID − FNL	+44.2 [+43.0, +45.4]	+24.9 [+22.5, +26.6]	+19.8 [+18.3, +23.5]
		STRONG	26.7 [26.1, 27.4]	94.6 [94.0, 95.1]	121.9 [119.9, 122.7]
		SOLID − STRONG	+44.1 [+42.8, +45.4]	+24.9 [+22.5, +26.6]	+20.3 [+18.8, +23.5]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	71.8 [70.8, 72.9]	120.3 [117.2, 122.7]	142.8 [141.8, 144.6]
		Jin	108.3 [108.1, 108.3]	151.7 [151.2, 152.1]	188.5 [188.0, 188.8]
		SOLID − Jin	−36.5 [−37.3, −35.2]	−31.4 [−34.1, −29.1]	−45.7 [−46.6, −43.9]
		ADE-BP	73.8 [71.5, 75.6]	120.4 [118.4, 123.0]	151.3 [147.2, 153.4]
		SOLID − ADE-BP	−2.0 [−3.6, +0.6]	−0.1 [−3.4, +3.4]	−8.5 [−11.4, −3.7]
		FNL	25.4 [25.1, 25.7]	94.9 [94.5, 95.3]	120.1 [119.4, 120.5]
		SOLID − FNL	+46.4 [+45.4, +47.5]	+25.3 [+22.4, +27.8]	+22.6 [+21.7, +24.8]
		STRONG	25.4 [25.2, 25.7]	94.9 [94.5, 95.3]	120.1 [119.4, 120.3]
		SOLID − STRONG	+46.4 [+45.4, +47.5]	+25.3 [+22.4, +27.8]	+22.7 [+21.7, +24.8]
		100.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-XII
FULL RESULTS OF BREGAR IN ANC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
3 or more anchors	1	SOLID	20.2 [13.0, 21.6]	21.9 [21.2, 24.6]	25.7 [21.8, 26.4]
		Jin	19.7 [15.9, 23.5]	24.2 [22.5, 28.2]	29.0 [24.3, 30.9]
		SOLID − Jin	+0.5 [−4.2, +2.9]	−2.4 [−6.3, +1.2]	−3.3 [−8.4, +1.1]
		ADE-BP	16.4 [11.9, 18.7]	19.1 [17.0, 20.7]	21.5 [19.1, 24.4]
		SOLID − ADE-BP	+3.8 [+0.5, +6.6]	+2.7 [+1.7, +5.2]	+4.2 [−0.7, +5.2]
		FNL	45.8 [40.2, 51.2]	53.4 [47.8, 87.6]	88.4 [53.4, 103.0]
		SOLID − FNL	−25.5 [−32.2, −20.7]	−31.6 [−65.3, −26.0]	−62.7 [−76.6, −27.7]
		STRONG	21.7 [16.1, 25.6]	26.5 [23.2, 32.5]	32.8 [26.5, 35.6]
		SOLID − STRONG	−1.5 [−6.3, +3.2]	−4.7 [−10.3, −0.9]	−7.1 [−13.2, −1.2]
		100.0 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	20.3 [12.5, 21.7]	22.4 [20.9, 25.1]	25.1 [21.8, 26.2]
		Jin	19.0 [15.9, 23.2]	23.6 [22.5, 28.4]	28.7 [23.6, 30.1]
		SOLID − Jin	+1.4 [−4.6, +2.9]	−1.2 [−5.3, +1.7]	−3.6 [−6.1, +1.6]
		ADE-BP	16.4 [11.4, 18.6]	19.1 [17.1, 21.0]	21.7 [19.1, 26.6]
		SOLID − ADE-BP	+3.9 [+0.9, +7.3]	+3.2 [+1.7, +5.1]	+3.4 [−0.7, +5.5]
		FNL	45.9 [39.3, 51.7]	51.9 [48.0, 87.9]	88.2 [51.9, 104.2]
		SOLID − FNL	−25.6 [−32.4, −19.7]	−29.5 [−64.8, −25.8]	−63.1 [−78.0, −26.8]
		STRONG	21.6 [13.9, 25.7]	26.2 [22.2, 31.8]	32.4 [26.2, 36.4]
		SOLID − STRONG	−1.3 [−5.9, +5.4]	−3.8 [−8.8, −0.2]	−7.3 [−12.9, −0.9]
		100.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-XIII
FULL RESULTS OF IDLAB IN AF

Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
1	SOLID	207.8 [176.4, 258.4]	376.0 [258.7, 568.0]	639.8 [511.3, 705.9]
	Jin	246.6 [162.3, 331.5]	588.7 [333.7, 617.9]	800.9 [603.4, 912.6]
	SOLID − Jin	−38.8 [−94.5, +22.2]	−212.6 [−337.0, +98.3]	−161.1 [−286.8, +72.3]
	FNL	277.5 [201.0, 317.0]	465.8 [319.3, 578.4]	692.3 [551.3, 734.1]
	SOLID − FNL	−69.8 [−86.1, −21.9]	−89.8 [−135.1, −3.0]	−52.5 [−100.1, +1.3]
	65.4 % of pairs and 100.0 % of scenarios scored			
10	SOLID	207.8 [169.4, 257.7]	374.6 [258.0, 545.4]	627.1 [507.8, 708.4]
	Jin	249.5 [163.4, 331.5]	588.8 [331.5, 614.9]	802.4 [600.4, 912.2]
	SOLID − Jin	−41.7 [−93.0, +21.2]	−214.1 [−341.4, +102.0]	−175.3 [−303.6, +58.3]
	FNL	279.8 [199.0, 322.0]	466.8 [317.3, 574.3]	691.3 [548.2, 731.1]
	SOLID − FNL	−72.0 [−92.6, −23.5]	−92.1 [−154.2, −6.5]	−64.2 [−113.5, +0.9]
	65.4 % of pairs and 100.0 % of scenarios scored			

TABLE S-XIV
FULL RESULTS OF IDLAB IN AAC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
All reporting cells	1	SOLID	205.5 [167.5, 245.1]	413.9 [361.5, 456.6]	679.4 [612.1, 731.7]
		Jin[a]	393.6 [344.3, 404.2]	494.9 [476.5, 536.3]	887.2 [748.3, 1007.2]
		SOLID − Jin	−188.2 [−224.5, −143.1]	−80.9 [−132.5, −44.4]	−207.8 [−362.0, −34.7]
		ADE-BP	279.0 [252.3, 300.6]	403.4 [375.7, 431.1]	604.8 [538.2, 678.2]
		SOLID − ADE-BP	−73.5 [−92.1, −52.1]	+10.5 [−31.3, +49.5]	+74.6 [−26.5, +168.8]
		FNL	233.9 [189.1, 269.3]	388.9 [354.3, 423.0]	543.6 [506.6, 577.0]
		SOLID − FNL	−28.4 [−45.6, −5.6]	+25.0 [−11.0, +57.7]	+135.8 [+73.6, +200.3]
		STRONG	282.2 [252.9, 308.5]	422.6 [393.9, 444.2]	562.4 [533.6, 599.8]
		SOLID − STRONG	−76.7 [−106.6, −42.7]	−8.6 [−49.2, +30.2]	+117.0 [+49.6, +183.5]
		67.6 % of pairs and 94.9 % of scenarios scored			
	10	SOLID	204.4 [166.6, 243.8]	413.7 [360.3, 454.2]	675.2 [610.0, 731.2]
		Jin[b]	393.2 [343.6, 404.2]	494.9 [476.5, 535.8]	887.0 [748.8, 1007.0]
		SOLID − Jin	−188.8 [−225.8, −144.1]	−81.2 [−133.8, −45.8]	−211.8 [−362.4, −36.9]
		ADE-BP	278.5 [252.1, 300.1]	402.6 [374.6, 430.5]	605.1 [538.0, 678.5]
		SOLID − ADE-BP	−74.1 [−93.0, −52.9]	+11.1 [−31.8, +48.6]	+70.1 [−29.8, +168.3]
		FNL	233.0 [187.8, 268.7]	387.0 [352.9, 421.4]	543.7 [504.7, 577.0]
		SOLID − FNL	−28.6 [−46.0, −6.1]	+26.7 [−11.2, +57.4]	+131.5 [+71.5, +200.9]
		STRONG	281.6 [253.9, 307.7]	419.4 [391.6, 440.4]	558.3 [531.7, 597.0]
		SOLID − STRONG	−77.2 [−108.2, −42.9]	−5.7 [−48.0, +32.1]	+116.9 [+50.1, +184.0]
		67.6 % of pairs and 94.9 % of scenarios scored			
1 anchor	1	SOLID	215.6 [180.5, 273.2]	420.1 [284.3, 551.2]	649.5 [527.6, 713.5]
		Jin	1717.8 [1344.8, 1760.2]	1780.5 [1772.5, 1800.9]	1827.9 [1823.4, 1836.6]
		SOLID − Jin	−1502.1 [−1551.1, −1158.4]	−1360.4 [−1492.9, −1233.0]	−1178.4 [−1300.3, −1119.7]
		ADE-BP	536.1 [368.6, 690.3]	861.0 [731.5, 1010.2]	1190.6 [990.2, 1315.4]
		SOLID − ADE-BP	−320.4 [−435.0, −184.0]	−441.0 [−640.2, −293.7]	−541.0 [−726.0, −348.8]
		FNL	275.4 [216.4, 320.5]	429.2 [341.2, 531.7]	613.6 [519.7, 716.1]
		SOLID − FNL	−59.8 [−75.1, −29.1]	−9.2 [−75.5, +50.7]	+36.0 [−32.3, +54.6]
		STRONG	326.4 [265.5, 362.2]	434.8 [376.2, 511.9]	603.3 [512.9, 662.1]
		SOLID − STRONG	−110.7 [−136.1, −59.6]	−14.7 [−97.3, +53.3]	+46.2 [−8.9, +76.4]
		65.4 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	213.8 [177.8, 272.8]	411.5 [285.5, 544.4]	636.0 [522.0, 703.8]
		Jin	1717.8 [1344.8, 1760.2]	1780.5 [1772.5, 1800.9]	1827.9 [1823.4, 1836.6]
		SOLID − Jin	−1504.0 [−1552.8, −1160.0]	−1369.0 [−1491.4, −1240.0]	−1191.9 [−1305.9, −1131.2]
		ADE-BP	532.3 [369.0, 690.4]	861.4 [732.1, 1006.1]	1192.1 [990.0, 1321.0]
		SOLID − ADE-BP	−318.6 [−436.9, −186.4]	−449.9 [−642.4, −298.4]	−556.1 [−737.3, −352.2]
		FNL	275.2 [216.7, 324.0]	426.4 [341.1, 525.3]	609.1 [517.7, 715.5]
		SOLID − FNL	−61.4 [−79.1, −30.1]	−14.9 [−76.4, +48.7]	+26.9 [−41.1, +50.4]
		STRONG	317.9 [263.8, 355.2]	425.3 [371.3, 514.1]	603.6 [517.6, 648.6]
		SOLID − STRONG	−104.1 [−135.5, −54.8]	−13.9 [−89.3, +60.6]	+32.3 [−14.8, +79.3]
		65.4 % of pairs and 100.0 % of scenarios scored			
2 anchors	1	SOLID	231.9 [189.1, 284.5]	422.7 [332.0, 505.6]	629.1 [557.1, 686.4]
		Jin	494.9 [457.5, 594.1]	940.7 [767.3, 1068.5]	1253.3 [1177.2, 1315.5]
		SOLID − Jin	−263.0 [−322.0, −238.7]	−517.9 [−634.1, −378.5]	−624.2 [−689.9, −556.9]
		ADE-BP	342.4 [297.3, 389.3]	616.6 [459.8, 714.8]	842.9 [733.6, 914.3]
		SOLID − ADE-BP	−110.5 [−132.8, −83.4]	−193.9 [−288.3, −73.5]	−213.7 [−297.5, −110.2]
		FNL	267.0 [222.1, 308.9]	410.4 [352.0, 473.2]	604.2 [533.7, 660.9]
		SOLID − FNL	−35.2 [−46.4, −18.7]	+12.4 [−24.5, +40.0]	+24.9 [+5.5, +46.0]
		STRONG	335.0 [282.0, 361.5]	441.2 [398.6, 510.3]	625.8 [568.4, 663.2]
		SOLID − STRONG	−103.2 [−127.6, −64.0]	−18.5 [−67.9, +9.8]	+3.3 [−24.9, +33.2]
		66.2 % of pairs and 100.0 % of scenarios scored			
	10	SOLID	230.1 [185.5, 283.7]	419.0 [328.3, 502.3]	628.5 [556.7, 685.8]
		Jin	494.9 [456.1, 595.2]	941.2 [768.4, 1067.5]	1252.3 [1176.2, 1317.3]
		SOLID − Jin	−264.8 [−324.6, −240.3]	−522.2 [−636.2, −381.8]	−623.8 [−688.3, −555.5]
		ADE-BP	340.9 [297.4, 387.8]	615.9 [453.7, 713.4]	840.6 [733.2, 914.6]
		SOLID − ADE-BP	−110.9 [−132.0, −84.5]	−196.9 [−290.0, −73.9]	−212.1 [−298.5, −110.7]
		FNL	266.6 [221.2, 308.3]	408.6 [350.7, 471.0]	604.0 [532.9, 660.1]
		SOLID − FNL	−36.5 [−47.5, −19.6]	+10.3 [−27.5, +39.5]	+24.5 [+3.2, +45.9]
		STRONG	330.5 [278.5, 360.5]	438.0 [397.4, 507.1]	622.4 [565.4, 657.8]
		SOLID − STRONG	−100.4 [−125.5, −62.2]	−19.0 [−72.0, +13.1]	+6.1 [−25.4, +35.5]
		66.2 % of pairs and 100.0 % of scenarios scored			

[a]Failed on 2 solves, 1 by a crash of the solver process. [b]Failed on 11 solves, 5 by a crash of the solver process.

TABLE S-XV
FULL RESULTS OF IDLAB IN ANC

Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
3 or more anchors	1	SOLID	310.1 [291.8, 327.4]	485.2 [460.8, 514.0]	685.5 [649.6, 732.1]
		Jin	358.2 [334.6, 378.5]	536.7 [512.5, 556.0]	688.6 [667.1, 704.5]
		SOLID − Jin	−48.0 [−63.3, −32.6]	−51.5 [−66.6, −30.6]	−3.1 [−33.0, +44.5]
		ADE-BP	365.4 [341.6, 388.7]	505.4 [489.6, 518.8]	633.1 [608.6, 659.8]
		SOLID − ADE-BP	−55.2 [−65.5, −44.9]	−20.2 [−38.3, +4.9]	+52.4 [+9.3, +109.2]
		FNL	384.3 [357.3, 412.6]	540.1 [521.2, 550.1]	677.5 [653.4, 689.0]
		SOLID − FNL	−74.1 [−91.8, −61.3]	−55.0 [−72.9, −28.5]	+8.0 [−28.9, +67.0]
		STRONG	384.7 [358.6, 414.5]	541.1 [522.6, 550.1]	678.4 [656.5, 691.0]
		SOLID − STRONG	−74.6 [−93.3, −62.4]	−55.9 [−74.8, −28.8]	+7.1 [−30.7, +64.7]
		40.9 % of pairs and 85.3 % of scenarios scored			
	10	SOLID	309.2 [290.6, 326.5]	484.2 [459.7, 512.9]	684.6 [648.0, 730.3]
		Jin[a]	357.6 [334.3, 377.7]	536.3 [511.9, 555.8]	687.3 [667.0, 703.1]
		SOLID − Jin	−48.4 [−63.7, −33.0]	−52.1 [−66.9, −31.6]	−2.7 [−34.0, +43.5]
		ADE-BP	365.0 [341.0, 388.6]	504.3 [488.9, 517.9]	632.3 [608.2, 657.7]
		SOLID − ADE-BP	−55.8 [−66.0, −45.2]	−20.1 [−38.7, +5.0]	+52.3 [+7.9, +107.9]
		FNL	383.4 [355.9, 410.7]	540.8 [520.0, 550.1]	678.5 [652.2, 687.9]
		SOLID − FNL	−74.2 [−90.4, −61.0]	−56.6 [−74.2, −29.2]	+6.1 [−30.1, +66.7]
		STRONG	381.8 [353.0, 409.2]	541.3 [518.0, 550.1]	676.0 [649.9, 685.8]
		SOLID − STRONG	−72.6 [−88.5, −59.5]	−57.1 [−73.9, −30.9]	+8.6 [−28.2, +68.5]
		40.9 % of pairs and 85.3 % of scenarios scored			
1 anchor	1	SOLID	409.6 [385.8, 433.8]	582.6 [555.4, 599.6]	667.4 [645.3, 692.3]
		Jin	1091.5 [1038.2, 1111.2]	1348.2 [1259.3, 1397.5]	1557.9 [1499.2, 1588.0]
		SOLID − Jin	−681.9 [−699.0, −649.7]	−765.6 [−807.4, −673.3]	−890.5 [−912.2, −849.4]
		ADE-BP	897.6 [829.7, 941.3]	1062.2 [1040.5, 1080.8]	1205.2 [1169.8, 1234.2]
		SOLID − ADE-BP	−488.0 [−514.1, −443.3]	−479.6 [−496.3, −466.2]	−537.8 [−575.0, −497.8]
		FNL	606.5 [564.7, 632.3]	775.7 [730.8, 823.0]	920.2 [882.6, 942.1]
		SOLID − FNL	−196.9 [−210.2, −176.4]	−193.0 [−229.2, −162.2]	−252.8 [−276.4, −219.1]
		STRONG	606.5 [564.7, 632.3]	775.7 [730.8, 823.0]	920.2 [882.6, 942.1]
		SOLID − STRONG	−196.9 [−210.2, −176.4]	−193.0 [−229.2, −162.2]	−252.8 [−276.4, −219.1]
		5.6 % of pairs and 59.7 % of scenarios scored			
	10	SOLID	409.9 [385.8, 434.9]	582.0 [552.3, 599.7]	666.0 [643.8, 690.8]
		Jin	1091.5 [1038.2, 1111.2]	1348.2 [1259.3, 1397.5]	1557.9 [1499.2, 1588.0]
		SOLID − Jin	−681.6 [−698.3, −649.6]	−766.2 [−807.7, −674.1]	−891.8 [−914.1, −850.6]
		ADE-BP	897.7 [828.8, 938.5]	1061.0 [1038.8, 1079.4]	1203.2 [1171.1, 1231.4]
		SOLID − ADE-BP	−487.8 [−511.6, −441.6]	−479.0 [−494.6, −465.4]	−537.2 [−575.2, −500.4]
		FNL	606.1 [563.7, 633.6]	774.6 [729.9, 821.4]	922.5 [880.8, 943.8]
		SOLID − FNL	−196.2 [−209.5, −175.3]	−192.6 [−229.5, −158.7]	−256.5 [−279.0, −218.1]
		STRONG	609.3 [564.3, 634.0]	779.0 [732.9, 824.6]	921.4 [883.5, 947.8]
		SOLID − STRONG	−199.4 [−212.2, −178.8]	−197.0 [−231.1, −166.6]	−255.4 [−280.7, −220.3]
		5.6 % of pairs and 59.7 % of scenarios scored			
2 anchors	1	SOLID	493.3 [465.6, 527.6]	734.3 [694.3, 779.5]	934.5 [901.2, 1056.9]
		Jin	689.0 [638.4, 721.2]	913.2 [874.2, 942.7]	1131.2 [1110.3, 1158.9]
		SOLID − Jin	−195.7 [−224.7, −152.5]	−178.9 [−236.8, −111.6]	−196.7 [−246.3, −66.2]
		ADE-BP	596.5 [557.1, 647.3]	806.4 [753.5, 843.6]	972.7 [942.3, 996.9]
		SOLID − ADE-BP	−103.3 [−155.5, −49.1]	−72.1 [−138.4, +6.9]	−38.2 [−86.1, +96.9]
		FNL	531.5 [519.4, 544.2]	679.8 [648.7, 696.4]	830.6 [791.6, 849.0]
		SOLID − FNL	−38.2 [−64.7, −4.7]	+54.5 [+6.8, +118.2]	+103.9 [+60.4, +238.0]
		STRONG	531.3 [519.1, 543.7]	679.7 [648.5, 695.8]	830.6 [791.5, 848.9]
		SOLID − STRONG	−38.0 [−64.7, −4.2]	+54.6 [+7.1, +118.2]	+103.9 [+60.6, +238.3]
		15.2 % of pairs and 75.5 % of scenarios scored			
	10	SOLID	493.3 [464.8, 526.9]	729.8 [693.8, 779.1]	935.3 [901.4, 1053.0]
		Jin	688.1 [637.7, 721.8]	910.7 [876.8, 942.4]	1131.5 [1110.0, 1159.2]
		SOLID − Jin	−194.8 [−224.6, −151.2]	−180.9 [−238.2, −113.3]	−196.2 [−245.7, −66.7]
		ADE-BP	595.7 [556.1, 645.7]	805.5 [752.2, 844.8]	973.1 [942.6, 996.7]
		SOLID − ADE-BP	−102.3 [−154.3, −51.0]	−75.8 [−140.1, +6.6]	−37.9 [−85.1, +90.4]
		FNL	530.0 [519.2, 544.4]	680.1 [648.4, 695.8]	829.9 [791.2, 847.6]
		SOLID − FNL	−36.7 [−65.0, −5.2]	+49.7 [+6.1, +116.1]	+105.4 [+61.5, +234.8]
		STRONG	530.8 [518.5, 545.3]	679.6 [647.2, 694.9]	830.1 [790.2, 846.6]
		SOLID − STRONG	−37.5 [−65.0, −5.0]	+50.2 [+6.1, +116.6]	+105.1 [+62.0, +236.2]
		15.2 % of pairs and 75.5 % of scenarios scored			

[a]Failed on 11 solves, 2 by a crash of the solver process.

TABLE S-XVI
RESULTS UNDER THE RANGED PAIRS WHERE THEY DIFFER FROM THE JOINED PAIRS

Dataset	Anchors	Readings	Method or difference	S50 (cm)	S75 (cm)	S90 (cm)
HYMN	1 anchor	1	SOLID	1034.2 [908.3, 1145.1]	1273.1 [1185.7, 1427.7]	1624.8 [1421.1, 1837.4]
			Jin	1756.5 [1694.9, 1819.4]	1893.2 [1829.4, 1939.4]	1980.4 [1934.8, 2182.9]
			SOLID − Jin	−722.2 [−855.2, −564.1]	−620.0 [−710.4, −416.6]	−355.5 [−679.2, −141.7]
			ADE-BP	1335.9 [1237.7, 1370.9]	1478.5 [1396.8, 1545.6]	1609.5 [1534.9, 1819.0]
			SOLID − ADE-BP	−301.7 [−431.9, −133.8]	−205.4 [−318.9, −16.8]	+15.3 [−389.1, +267.5]
			FNL	487.6 [391.2, 683.7]	899.4 [683.7, 1022.8]	1065.7 [959.2, 1344.2]
			SOLID − FNL	+546.7 [+263.1, +715.4]	+373.8 [+202.9, +737.4]	+559.1 [+208.2, +814.5]
			STRONG	487.6 [391.2, 683.7]	899.4 [683.7, 1022.8]	1065.7 [959.2, 1344.2]
			SOLID − STRONG	+546.7 [+263.1, +715.4]	+373.8 [+202.9, +737.4]	+559.1 [+208.2, +814.5]
			87.1 % of pairs and 100.0 % of scenarios scored			
AAU	3 or more anchors	1	SOLID	5.1 [4.8, 5.4]	11.1 [10.1, 12.0]	72.2 [61.0, 84.7]
			Jin	7.7 [7.2, 8.3]	52.3 [45.7, 59.3]	121.8 [114.6, 129.6]
			SOLID − Jin	−2.6 [−3.3, −2.1]	−41.3 [−47.7, −35.2]	−49.6 [−58.9, −39.2]
			ADE-BP	24.8 [23.9, 25.7]	44.1 [41.9, 46.5]	95.1 [87.3, 105.9]
			SOLID − ADE-BP	−19.7 [−20.7, −18.8]	−33.0 [−35.2, −31.1]	−22.8 [−31.8, −13.6]
			FNL	4.3 [4.0, 4.7]	31.0 [27.4, 33.9]	124.4 [110.7, 137.3]
			SOLID − FNL	+0.8 [+0.5, +1.0]	−19.9 [−22.3, −16.8]	−52.2 [−63.0, −41.3]
			STRONG	25.0 [24.1, 26.3]	47.5 [44.6, 51.2]	125.1 [111.4, 138.9]
			SOLID − STRONG	−19.9 [−21.2, −19.0]	−36.4 [−39.7, −33.8]	−52.8 [−63.8, −42.6]
			98.6 % of pairs and 99.8 % of scenarios scored			
AAU	1 anchor	1	SOLID	277.7 [271.2, 282.4]	302.4 [300.0, 305.0]	319.6 [316.3, 322.4]
			Jin	473.0 [473.0, 476.2]	493.5 [492.5, 494.8]	499.7 [499.7, 502.2]
			SOLID − Jin	−195.3 [−203.2, −190.6]	−191.1 [−195.1, −187.9]	−180.1 [−185.5, −177.6]
			ADE-BP	384.3 [380.8, 386.9]	412.6 [408.7, 414.7]	431.0 [427.6, 435.1]
			SOLID − ADE-BP	−106.6 [−113.7, −100.8]	−110.2 [−114.1, −105.6]	−111.3 [−117.1, −106.5]
			FNL	241.8 [241.4, 243.6]	303.3 [302.9, 304.1]	377.5 [375.0, 388.8]
			SOLID − FNL	+35.9 [+29.2, +40.6]	−1.0 [−3.7, +1.7]	−57.8 [−69.7, −53.3]
			STRONG	241.8 [241.4, 243.6]	303.3 [302.9, 304.1]	377.5 [375.0, 388.8]
			SOLID − STRONG	+35.9 [+29.2, +40.6]	−1.0 [−3.7, +1.7]	−57.8 [−69.7, −53.3]
			86.8 % of pairs and 100.0 % of scenarios scored			
AAU	2 anchors	1	SOLID	167.4 [161.2, 173.2]	202.3 [198.6, 206.9]	285.7 [257.7, 313.9]
			Jin	215.6 [210.5, 221.0]	279.2 [274.2, 283.3]	363.3 [354.0, 375.9]
			SOLID − Jin	−48.3 [−53.2, −43.9]	−76.9 [−81.3, −69.3]	−77.6 [−107.1, −50.0]
			ADE-BP	185.2 [180.2, 190.3]	224.9 [218.4, 230.4]	275.7 [265.5, 288.1]
			SOLID − ADE-BP	−17.8 [−23.5, −12.2]	−22.6 [−27.7, −16.4]	+10.0 [−16.4, +35.5]
			FNL	145.3 [140.5, 153.9]	213.6 [206.1, 218.8]	256.6 [247.5, 270.9]
			SOLID − FNL	+22.1 [+11.6, +29.2]	−11.3 [−17.8, −2.2]	+29.0 [−2.4, +59.9]
			STRONG	145.3 [140.5, 153.9]	213.6 [206.1, 218.8]	255.8 [247.5, 270.9]
			SOLID − STRONG	+22.1 [+11.5, +29.2]	−11.3 [−17.8, −2.2]	+29.9 [−2.6, +60.4]
			97.0 % of pairs and 100.0 % of scenarios scored			

TABLE S-XVII
S75 ON THE TRAINING PARTS WITH ONE PART OF THE BLOCK LIKELIHOOD REMOVED

Dataset	Mode	Readings	SOLID	No heavy tail	No excess	No offset	Best baseline of Table II
TWR-CLOUD	AF	1	76.0	206.8 (6.1)	81.9 (2.7)	76.4 (2.3)	FNL 270.7
		10	57.9	187.1 (7.1)	65.6 (3.3)	58.3 (2.2)	FNL 70.7
	AAC	1	18.6	159.3 (3.8)	26.8 (0.8)	19.1 (0.5)	ADE-BP 30.3
		10	16.2	123.6 (4.6)	25.5 (1.3)	25.4 (1.3)	ADE-BP 26.4
	ANC	1	62.4	184.8 (5.4)	67.2 (1.7)	60.9 (0.5)	ADE-BP 89.3
		10	36.2	137.0 (6.2)	44.1 (2.3)	43.7 (1.9)	Jin 64.5
TIERS	AF	1	7.3	7.3 (0.0)	10.2 (0.2)	7.6 (0.2)	Jin 9.7
		10	7.2	7.2 (0.0)	10.4 (0.0)	9.1 (0.2)	Jin 10.0
	AAC	1	7.5	7.5 (0.0)	11.5 (0.1)	8.2 (0.1)	ADE-BP 8.2
		10	7.3	7.3 (0.0)	11.4 (0.0)	9.3 (0.1)	ADE-BP 8.1
	ANC	1	13.2	13.2 (0.0)	19.5 (0.1)	13.5 (0.1)	ADE-BP 12.4
		10	13.0	13.0 (0.0)	19.2 (0.1)	13.8 (0.0)	ADE-BP 12.2
Bregar	ANC	1	22.7	23.5 (0.3)	58.8 (12.4)	19.2 (1.5)	ADE-BP 18.9
		10	22.6	22.6 (0.1)	59.6 (12.3)	25.5 (2.2)	ADE-BP 19.2
IDLab	AF	1	484.2	485.5 (11.2)	489.8 (12.5)	480.6 (9.4)	FNL 534.3
		10	482.9	484.8 (5.8)	489.5 (9.9)	474.0 (12.1)	FNL 534.0
	AAC	1	422.3	422.7 (0.8)	425.9 (1.8)	422.6 (1.0)	FNL 406.3
		10	421.0	425.3 (2.1)	422.8 (1.8)	421.7 (2.4)	FNL 402.8
	ANC	1	482.6	481.6 (1.1)	484.4 (1.2)	483.3 (1.3)	ADE-BP –
		10	481.7	482.7 (1.4)	483.4 (1.3)	482.3 (1.1)	ADE-BP –

S75 in cm, in ANC over three or more anchors, with the standard error of the difference from SOLID in parentheses and a dash where no full run used the evaluated options.

S-IV. The Block Likelihood of One TWR-CLOUD Link

The link of Section I, whose block likelihood Section IV-A discusses, joins agent 85 and anchor 80 in scenario 603054 of the sixth training scan of TWR-CLOUD. Its ten readings agreed within 1.8 cm and lay 35.8 to 37.5 cm above its true distance of 349.3 cm. Fig. S6 plots its log block likelihood (6) at the fitted values of TWR-CLOUD (Table III) against the distance, less its peak at 385.0 cm. The true distance lies 2.3 nats below that peak, and 422.1 cm, as far above the median reading as the true distance lies below it, 5.7 nats below. An excess explains readings that run long but none that run short, so the link yields to a shorter distance more readily than to a longer one (Section IV-A).

S-V. Details of the Comparison

This section completes the comparison of Section VII-B, while the details of each dataset stand in Section S-II.

A. The Baselines

We implemented the four baselines from their papers and checked each implementation against the equations and the stated numerical procedure of its paper, and the semidefinite programs of Jin et al. also against the code that the authors published. Jin et al. [12] solve one semidefinite program for the least residual and a second one for the sparse nonnegative link biases, under the residual bound of their paper, twice that least residual. We solve both with the Clarabel interior-point solver in place of the SeDuMi solver of their released code. Both programs hold the positions and their inner products in one matrix variable, the lifted matrix. In AF, which their paper does not treat, we read the coordinates of the agents from the centered block of their inner products. In AAC and ANC the published recovery reads the coordinates from the cross block of the lifted matrix. It leaves every agent at the anchor when a scenario has one anchor and loses the coordinate across the line when all anchors lie on one line. We score these estimates as the published recovery returns them, which explains the large AAC errors of Jin et al. with one and two anchors on TIERS and IDLab (Tables S-X and S-XIV).

ADE-BP [13] runs belief propagation with the empirical asymmetric double exponential law of the ranging error of Li et al., model 1 with constant scales or model 2 with scales linear in the reading. It keeps the 500 particles and, in ANC, the four rounds of message passing in which it converged in its paper. Its paper asks the inner importance refinement to converge but states no numerical rule that ends it, so a refinement ends once the fitted mean and covariance stay stable over two consecutive steps, or after 256 refinements on every dataset. Stable means a change of the mean below 0.001 of the diameter of the search area and a change of the covariance below 0.0001 of its square. On the training scans of TWR-CLOUD, in its evaluated configuration, the caps 512 and 1024 raised the mean share of its converged solves over AAC and ANC from 0.62 to 0.65 and 0.67 and lowered the mean of its S75 over these two modes and the two reading counts by 0.06 and 0.05 cm, so a larger cap would barely change its accuracy. Its paper treats no AF and leaves heights to a remark that a three-dimensional extension is straightforward, so it runs only in the modes with anchors and keeps its planar form on Bregar, HYMN, and AAU, although their radios spread in height. On the training parts of Bregar, model 2 widens its long side by 0.068 to 0.077 times the distance, and in environments 0, 2, and 3 it gave ADE-BP a mean S75 over the two reading counts 2.7 to 4.0 cm below that of model 1, while in environment 1 the two lay within 0.2 cm of each other and ADE-BP uses model 1.

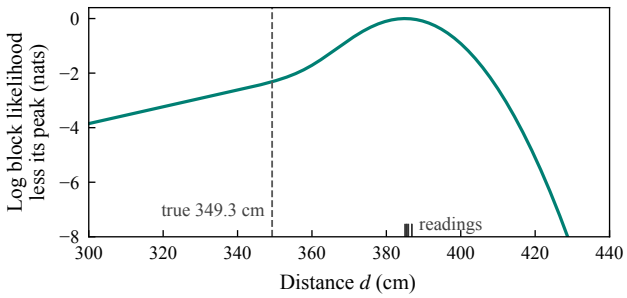


Fig. S6. Block likelihood of the TWR-CLOUD link of Section I.

FNL of Gur and Sabach [14] keeps its published schedule of 10 000 updates and its step size, and it weights each reading by the inverse square of its reported scale. Its paper assumes at least one anchor, so in AF we spread the agents of a component that start at one point evenly over a circle about the origin whose radius is half the longest reading between them. STRONG of Soares and Gomes [15] keeps its convex Huber underestimator and its accelerated solver, with a cap of 10 000 iterations and a relative stopping tolerance of 10^{-5}. Without anchors its underestimator reaches zero at the layout in which all agents coincide, so it does not run in AF.

The rule of Section VII-B averages the balanced S75 under the joined pairs, over the scenarios with three or more anchors in ANC and over all scenarios in the other modes, counts means within 0.01 cm of the lowest as tied, and breaks a tie by a fixed order of the candidates. Jin et al. select their planar or spatial model and the median or the mean of the readings of a link. FNL selects its model, the median of a link or every reading as its own term, and its start (Section S-V-B). STRONG selects its Huber radius of 10, 20, 40, 80, or 160 cm or no robust loss, its model, its start, and the median of a link or every reading as its own term. ADE-BP selects its model, the median of a link or every reading as its own term, and in AAC four or ten rounds. Table S-XVIII lists the choices, with given or initializer as the start of FNL and a length as the Huber radius of STRONG. ADE-BP uses the median of a link wherever it runs, and on Bregar every baseline selects its options per environment.

B. The Inputs and Starts

The operating area that SOLID and ADE-BP both receive (Section VII-B) bounds the ANC posterior of SOLID and the search area of ADE-BP, and Section S-I-G gives its rules and those of the height prior that SOLID alone receives.

TABLE S-XVIII
OPTIONS OF THE BASELINES

Dataset	Mode	Jin	ADE-BP	FNL	STRONG
TWR-CLOUD	AF	planar, median	–	planar, median, initializer	–
	AAC	planar, median	model 2, 10 rounds	planar, median, initializer	planar, median, 40 cm
	ANC	planar, median	model 2, 4 rounds	spatial, median, given	planar, median, 20 cm
TIERS	AF	planar, median	–	planar, median, initializer	–
	AAC	planar, mean	model 2, 10 rounds	spatial, median, initializer	planar, every reading, 160 cm
	ANC	planar, median	model 2, 4 rounds	spatial, median, initializer	spatial, median, 160 cm
IDLab	AF	planar, mean	–	planar, median, initializer	–
	AAC	spatial, mean	model 2, 10 rounds	spatial, median, initializer	planar, every reading, no robust loss
	ANC	spatial, mean	model 1, 4 rounds	spatial, every reading, initializer	planar, every reading, 160 cm
Bregar	ANC	per environment	model 2, 4 rounds, environment 1 model 1	per environment, given	environment 0 spatial, every reading, 10 cm, environment 1 planar, median, 40 cm, environments 2 and 3 planar, median, 160 cm
HYMN	ANC	spatial, median	model 2, 4 rounds	planar, every reading, initializer	planar, median, 40 cm
AAU	ANC	spatial, median	model 2, 4 rounds	spatial, median, initializer	planar, median, 10 cm

A dash marks a mode in which a baseline does not run.

The cooperative modes of SOLID read no area. FNL and STRONG start from the given starting coordinates of each scenario (Section S-II) or from the initializer layout of SOLID (Section S-I-D), as their selection takes it, which is the initializer layout wherever STRONG runs and the start that Table S-XVIII gives for FNL. The initializer then reads each link as the baseline does, by its median or by every reading as a term of its own, and for STRONG it scores the laterations with δ equal to the Huber radius in centimeters, or 10^9 without a robust loss, in place of $\delta = 0.7$ (Table S-III). SOLID starts its cooperative modes from its own start layouts (Section S-I-D), while its ANC, the relaxation of Jin et al., and the particles of ADE-BP take no start.

C. Missing Coordinates, Failures, and Solves That Stayed at Their Start

A coordinate that is not finite or lies beyond 100 km is invalid and counts as missing for every method, and a solve that crashes its process counts as a failed solve of its method, each of whose pairs costs the cap of (1), as a missing coordinate does (Section VII-B). Jin et al. failed on one TWR-CLOUD solve and on 24 IDLab solves of the evaluation parts, 8 of them by a crash of their solver process, and with their selected options on 2 TWR-CLOUD and 60 IDLab solves of the training parts, 1 and 20 of them by a crash. In 2 more IDLab solves of the evaluation part and 3 of the training part, their layout held invalid coordinates. The selection rule scores a failure in the same way.

A solve stays at its start when its estimate equals the positions it starts from, agent for agent and to the last bit, so a solve that moves an agent by rounding alone counts as moved. The test applies to FNL and STRONG, the only methods that start from positions they receive (Section S-V-B). Table S-XIX counts these solves for every dataset and mode, and in ANC also for one and for two anchors where a dataset has such scenarios. On TIERS and AAU, FNL and STRONG both start ANC from the initializer layout of SOLID and use the median of a link (Table S-XVIII). With one anchor that layout places every agent with a link on one ray from the anchor, along the first coordinate axis, at the distance of its median reading (Section S-I-D), where that median fits exactly. It leaves every other agent at its starting coordinates (Section S-II), where no reading acts on it. Both therefore stayed at their start in every one-anchor scenario of the two datasets, as Section VIII reports.

TABLE S-XIX
SOLVES OF FNL AND STRONG THAT STAYED AT THEIR START

Dataset	Mode	All solves	FNL	STRONG
TWR-CLOUD	AF	98 840	0	–
	AAC	494 200	0	121
	ANC	494 200	0	86 298
TIERS	AF	140	0	–
	AAC	34 440	0	39
	ANC	34 440	1168	2094
	1 anchor	1120	1120	1120
	2 anchors	3920	48	748
IDLab	AF	5416	0	–
	AAC	1 056 120	181 976	152 245
	ANC	1 056 120	266 980	333 823
	1 anchor	81 240	47 415	47 784
	2 anchors	81 240	32 979	41 005
Bregar	ANC	96	0	0
HYMN	ANC	1200	57	209
	1 anchor	120	57	119
	2 anchors	120	0	80
AAU	ANC	10 458	1497	3236
	1 anchor	1494	1494	1494
	2 anchors	1494	3	1339

A dash marks a mode in which STRONG does not run.

The controlled timing on TWR-CLOUD (Section S-III) also records how every solve ended. ADE-BP reached its cap of refinements in some update of 139 162 of its 494 200 solves of AAC and of 150 800 of those of ANC, and FNL ended every solve at its fixed budget of updates. Every run of the block stage of SOLID converged in 93 319 of its 98 840 solves of AF and in 493 593 of its 494 200 solves of AAC. In the others some run ended at the cap of 10^4 iterations or without an accepted step, which the line search of Section S-I-D allows.

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